FITTER-EU

DELIVERABLE D2.2

Embedding social justice in twin transition research: Concepts, definitions and methodological approach (2)

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Summary

D2.2 presents an analysis of the meaning of the twin transition and further elaborates the research methodology to be undertaken in FITTER. D2.2 will be updated in M21.

Contents

1. Introduction

1.1. Overview of FITTER-EU

FITTER-EU aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem that includes a Digital Platform. Through a co-creation methodological approach, this ecosystem aims to enable policymakers to understand the structures and systems through which social groups risk being adversely affected by twin transition policies. This understanding is supported through the co-creation of different future scenarios, and the development of mitigation plans with the support of Better Practice Guides.

1.2. Purpose and scope of the Deliverable

This deliverable provides conceptual background for the FITTER understanding of just transition processes and contributes to existing knowledge by critically assessing mainstream approaches to the twin transition. The aim is to broaden our understanding of the relationship between the twin transition and existing structural inequalities through the problematization of how the economy, the labour market and inequalities themselves are defined in the European Union (EU) context, and through a holistic approach to policy making. This approach informs the development of the FITTER approach and FITTER methodology, providing critical lenses through which the other methodological steps of the research process will be examined.

1.3. Structure of the Deliverable

The report builds upon D2.1 in several ways. Chapter 2 provides an overview of existing inequalities in Europe, with the aim to identify potential structurally vulnerable groups. A summary of main themes related to twin transition, green transition and digital transformations is also included. This is followed by an overview of the transition in the four FITTER sectors of analysis - energy; construction and built environment; mobility and transport; and food and agriculture – which are discussed from the perspectives of key actors, with power relations, employment relations, and consumer perspectives discussed in each sector. Chapter 3 explores framings of sustainability, technological progress and the economy in more detail to further clarify the systems and structures that produce inequality and environmental harm. Chapter 4 describes how the methodology of the project is informed by this conceptual development.

1.4. Abbreviations and Acronyms

BAU	Business As Usual
CAP	Common Agricultural Policy
CCUS	Carbon capture, utilization and sequestration
EGD	European Green Deal
EU	European Union

FEC	Foundational Economy Collective
GDP	Gross Domestic Product
GHG	Green House Gas
GVA	Gross Value Added
ICT	Information and Communication Industry
ILO	International Labour Organisation
LGBTQI	Lesbian, gay, bisexual, transgender and queer people
OECD	The Organisation for Economic Co-operation and Development
SME	Small and Medium Enterprise
UN	United Nations
UNRISD	United Nations Research Institute for Social Development

2. Inequalities and twin transition in Europe: review of literature

The twin transition became a focus of EU policies recently, with increased emphasis on how digital technologies could assist and enable the green transition. Yet, the debate on how the two transitions should be combined is ongoing, with some criticism showing that the digital transition is not the answer to all challenges presented by the climate crisis.

There are also concerns regarding policies developed as part of the twin transition in Europe and their impact on structurally vulnerable groups. While the rhetoric of a 'just transition', defined as 'leaving no-one behind', runs in the background of most green policy development, it has been noted that this simplified definition needs to be unpacked in the context of how the two transitions are conceptualised, which groups are considered as structurally vulnerable, and how both translate into the concrete sectors of the economy.

This chapter aims to answer these questions. It starts with an overview of existing structural inequalities in contemporary EU, with more details on FITTER case-study countries provided in D2.1 Chapter 3. This is followed by a brief exploration of the academic debate regarding the twin transition, with a particular focus on structural inequalities. The final section provides a discussion on the twin transition and structural inequalities in the four FITTER sectors: energy, construction and built environment, transport, and agriculture. This sectoral review is conducted on the European level, with a focus on both employment in the sector and consumers, building on a more country-specific analysis of the four sectors provided in D2.1 Chapter 5. This will provide the background for the subsequent chapters, which propose a more detailed FITTER approach to a just twin transition.

2.1. Existing Inequalities in Europe: understanding the baseline

The twin transition takes place in societies that are already challenged by multiple intersecting inequalities. These need to be considered if transition policies aim to be inclusive, 'leaving no one behind', rather than exacerbating existing inequalities or creating new ones. While this document does not aim at providing a comprehensive analysis of inequalities in Europe, there are certain patterns which require to be mapped prior to considering potential inequality impacts of the twin transition on different social groups. First, the most pronounced structural inequalities within the nation states should be noted, as social structures produce systems of privilege and oppression/discrimination based on factors such as gender, socio-economic background, family status, ethnicity, migration background, and age, creating an unequal distribution of material resources, capabilities, and outcomes within or across societies and populations. Second, the section considers the ways in which state responses to macro-structural events (such as global financial crises or COVID-19 pandemic) impact structurally vulnerable cohorts[1]. Third, the intersectional and structural nature of inequalities is also discussed, to provide background for the FITTER approach, which will be discussed in detail in Chapter 4.

[1] A more detailed overview is provided in D2.1 Chapter 3, which also contains examples from the FITTER six national case studies

2.1.1. Structural inequalities and vulnerabilities: main inequality grounds in the context of recent socio-economic shifts

EU equality law covers discrimination grounds based on sex/gender, race and ethnicity, age, sexual orientation, disability, religion or belief. On the other hand, there is no agreed upon definition of vulnerability, nor an agreed upon criteria for identifying vulnerable groups, or a standard list of such groups in EU law or UN Human Rights Law. Typically, human rights bodies deal with vulnerable and disadvantaged communities on an ad hoc basis. (Chapman and Carbonetti 2011). In the FITTER context, vulnerability is defined as a multidimensional, differential, and dynamic concept, intimately linked with the persons' individual characteristics and experiences in a certain environment, assessed in relation to 3 main dimensions and their interconnections: labour market, financial resources and skills (European Commission, 2021).

Income inequality, poverty, and social exclusion and how they affect different groups is a cross-cutting theme when examining inequality and structural vulnerability. In 2023, 21.4% of people in the EU were at risk of poverty and social exclusion with multiple characteristics contributing to these, including household's level of income, joblessness, low work intensity or working status (Eurostat, 2024a). Some categories of structurally vulnerable people are also at higher risk of poverty and social exclusion, including single parents, younger persons, older citizens, members of ethnic minorities, among others (Copeland, 2023). Other factors, such as affordable housing, access to childcare or location (urban or rural) also have an impact (Duffy, 2020).

This can be further linked to **socio-economic background,** otherwise referred to as 'social origin' (Boke, 2022). Various grounds related to socio-economic discrimination are included in national legislation in several EU Member States, however the use of it varies from state to state (IHREC, 2024). Nevertheless, it has been recognized by numerous studies that, while socio-economic status often intersects with other inequalities, it is also a cause of social exclusion, preventing people from fully accessing employment, education or housing (Gutterman, 2023; Ridgeway, 2013. When measured in economic terms, socio-economic background hinders social mobility, as research shows. For example, in the European OECD countries, children from disadvantaged background earn on average 20% less than those with more favourable childhoods, while the socio-economic standing of parents also influences children's educational outcomes, leading to differences in job quality in later life (OECD, 2019). Socio-economic status can also impact on one's position during macro-economic shifts, such as the COVID-19 pandemic or the 2008 financial crisis, especially when combined with labour market position (Smith et al., 2023).

Gender remains one of the most pronounced inequality dimensions in Europe, as traditional gender roles, in particular those related to caregiving, are still deeply rooted in social norms, thus influencing women's paid and unpaid work (Eurofound, 2022b). Inequalities between men and women continue to be an issue – with substantial gender pay gaps, gaps in employment rates, disparities in political representation and representation in senior management as substantial points in the discussion on gender equality (EIGE, 2024). Gender inequalities also continue to persist in relation to care work and housework, both formal and informal, as women are overrepresented among care workers and domestic helpers (Wiese and Culot, 2022) while the bulk of housework is also done by women (EIGE, 2020). This can be also connected with a historical dualism between ‘productive’ and ‘unproductive work’, with men considered ‘economically productive while women who look after home full-time perceived as non-working (EIGE, 2021). Inequalities related to care work also intersect with other characteristics, most notably migration background, as care workers are overrepresented by female migrants (Wiese and Culot, 2022) who are also a subject of exploitation, especially if engaging in work in the informal economy (Fleury, 2019).

Inequalities related to gender have been also extensively discussed in the context of the COVID-19 pandemic and the state responses to the crises. As evidenced by various studies, women were more likely become unemployed as they were overrepresented in sectors affected by lockdowns and they were taking on disproportionally more caring duties during periods of restrictions (e.g., Bloom and Dobrotic; Fodor et al., 2021; Grasso et al., 2021). In the longer-term, women also tend to be more impacted by state responses to economic downturns in general, as austerity measures usually lead to cuts in public services accessed mainly by women (Cook and Grimshaw, 2021).

Age is another important dimension of inequality, understood as inequalities linked with old age but also considered as a factor across the life course, with socio-economic analysis also often turning attention to issues such as child poverty, or the situation of young people. Examples include economic inequalities linked to old age, particularly in the light of a shift towards the privatisation of pensions (Ebbinghaus, 2017), and low pay work among older citizens. In this context, age also intersects with gender as women continue to have lower pensions than men, due to the gender pay gap, labour market participation (often linked with gender division of care work) and their higher share of part-time work when compared to men’s (Burkevica et al., 2015). On the other end of the spectrum, generational challenges affect young people in relation to their labour market position when entering the workforce during downturns (Anders and Macmillan, 2020), or their housing tenure in the context of the widespread housing crises. (Eurofound, 2024). In terms of the impact of macro-economic shifts, younger people were more likely to be impacted by the 2008 financial crash, for example, with long-term effects linked to the ‘scarring effect’ (Anders and Macmillan, 2020), which refers to the weaker labour market performance of younger cohorts who enter employment at the time of downturn. In terms of state responses to crises and inequalities related to age groups, it should be noted that older citizens were disproportionately affected by lockdown restrictions and other mitigation measures during the COVID-19 pandemic. In particular, their quality of life and access to care was profoundly impacted, and the measures resulted in social isolation and increased deterioration of their mental health (Eurofound, 2022a). Younger groups were also a vulnerable group as they were more likely to lose their jobs, suffer from financial insecurity and mental health problems (Eurofound, 2021).

There is also a strong correlation between **migration background** and inequalities. In Europe, migrants are facing relatively higher risk of social exclusion and poverty, especially if they originally came from a non-EU country (Eurostat, 2024b). Migrants are often placed in low pay sectors of the host country labour markets (Castles and Miller, 2015), excluded from political participation and segregated by the mode of entry which determines their civil and social rights (Taket et al., 2009). Yet, even though migrants are often absent from the decision making process, they are a subject of state policies. The most striking examples of the impact of such policies was the COVID-19 crisis: in various EU countries, migrants were not treated as full-citizens and thus were excluded from measures aimed at offsetting the impact of lockdowns, which often resulted in further marginalisation (Bobek and Sandstrom, 2024).

While socio-economic background, gender, age and migration status tend to be the most pronounced inequality grounds and are often considered by cross-national studies, it should also be emphasised that other factors would also require further examination in the context of the twin transition such as, for example, **disability**, especially considering the hidden and invisible nature of many disabilities. **Location** is also a factor to be considered. As will be discussed in the next section, spatial variations are important for the analysis at the European level, particularly in relation to variations between countries and between regions within nation states. In addition, **rural-urban** divisions related to households should also be explored as illustrative of structural inequalities. For example, in Ireland rural communities have been disproportionally affected by the two previous crises (Faulkner et al., 2019).

2.1.2. Understanding of structural inequalities: intersectionality, power relations and the limitations of available quantitative data.

The overview of the existing inequalities presented in the previous section is mainly based on quantitative analysis and often relies on availability of data from large national surveys. While it provides a bird-eye view, it has limitations: first, it tends to focus on one dimension of inequality; second, its focus is on the outcomes, rather than the causes of such inequalities. In relation to the first point, unidimensional approaches to the analysis of inequalities have been criticized, especially by feminist theorists, with the concept of intersectionality being proposed as a more appropriate tool for analysing inequalities. First coined by Crenshaw (1991) when referring to a legal case in which a black women claimed discrimination on sex and racial dimension simultaneously, intersectionality evolved through different strands of feminism theorising, including ecofeminism and post-structural feminism (Kajser and Kronsell, 2014). According to this approach, inequality indicators should not be treated separately or in an additive way but should rather be scrutinized as crosscutting (Gender Action, 2024). For example, gender intersects with other inequality factors, such as migration background, age or family status. Studies show that female migrants face multiple vulnerabilities, particularly in the context of migration and care work and domestic work (Fleury, 2016). Family status also intersects with gender in terms of structural inequalities – in the EU context, the majority of single-parent households are headed by women, and these households are also at risk of poverty and social exclusion (Nieuwenhuis, 2020).

Although intersectionality has gained more popularity in the mainstream social sciences in recent years, some gaps remain. As argued by Debusscher and Luna Maes (2024), more attention needs to be paid to how intersectionality is taken up in EU policy, particularly in the context of building a ‘Union of Equality’ which ensures the conditions for ‘everyone to live, thrive, and lead regardless of differences based on sex, racial or ethnic origin, religion or belief, disability, age or sexual orientation’ (European Commission, 2020a). Considering intersectional inequality in policy making is especially important given that these inequalities can be reinforced through laws or policy measures, with important intersections of power and dominance (a detailed discussion on this topic is presented in D2.1. Chapter 2). In relation to state responses to crises and their impact on intersecting inequalities, examples can be provided from the recent COVID-19 pandemic. As demonstrated by various studies, women were generally more affected by the mitigation measures and related restrictions, yet not all were impacted in the same way, as the negative effects were more severe on racialised and disabled women (Hankivsky and Kapilashrami, 2020; Molenaar, 2021).

In relation to the second issue (focus on outcomes rather than causes of inequalities), it should be noted that, to fully understand the ways in which we can transform unequal relationships, inequalities also need to be understood as *structural* rather than *natural.* In the context of macro-economic shifts, experiences from previous crisis (such as the Great Recession) show that households are increasingly required to provide their own safety nets during the times of economic downturns (Bryan and Rafferty, 2014), particularly in the light of eroding welfare state (Crouch, 2009). This issue is closely related to power dynamics, social institutions, and the ways in which they reproduce social stratification. An example is the way in which property ownership is structured in the context of paying the rent: privileges related to the possession (e.g. owing a property) allow the 'owner' to extract income from a disadvantaged individual, even though the activity is not necessarily productive (Sorensen, 1996). In that sense, inequalities cannot be taken out of context, and the dynamics of privilege should also be scrutinized. As noted earlier, this will be considered in the FITTER approach, further discussed in Chapter 4 of this report.

2.2. Twin transition: overview of the existing literature with focus on structural inequalities

The twin transition is now high on the policy agenda of the EU, with an emphasis in combining the green transition with the digital transition (Terzi et al., 2023). However, because this focus of policymaking is relatively new, there are significant gaps in the knowledge of the twin transition's societal impacts (Faggian et al., 2024), especially considering how 'policy is running ahead of the evidence' (Kovacic, 2024). A central question is whether the green and the digital transitions will go hand in hand and create synergies, or whether the twin transition will be unbalanced due to tensions between policies aimed at achieving either digital or green goals (Fouquet and Hippe, 2022). There is a possible complementarity between, for example, digitalisation and energy efficiency, with the former positively impacting the latter if specific policy frameworks are applied consistently (Hedberg and Sipka, 2020). However, some studies have shown that introducing advanced digital technologies does not always lead to more sustainability (Best et al, 2020) and that the evidence of firms being able to utilise digital technologies to become more environmentally friendly is limited and fragmented (Faggian et al., 2024). This contradiction of technology supporting the green transition and being counterproductive at the same time can be exemplified by remote work, which is facilitated by digitalization while reducing carbon emissions related to transport, yet also resulting in higher emissions due to increased residential energy use, and increased production, use and disposal of ICT (Lange et al 2020, Andrae and Edler, 2015; Belkhir and Elmeligi, 2018; Malmodin and Lundén, 2018). In this case, it has been argued that the potential conflicts and competing objectives between the two transitions need to be taken into account (Gao, 2024). In sum, taking for granted the digital transition as a 'magic bullet' for sustainability is problematic and, as highlighted by some scholars, the relationship should rather be treated as complex, with the digital sector perceived as both a solution to decarbonisation and a source of increased carbon footprint (Fazio et al., 2024).

On the other hand, as argued by Gao (2024), green and digital policies are often dealt with separately on the operational level, with fragmented governance and the under responsibility of different institutions. This poses a challenge when assessing the societal impacts of the twin transition with a view to making it inclusive and 'just'. Considering that the developments in the shift towards a more complex understanding of the dual nature of the transition are only recent, the following section explores the green and the digital transitions separately, with a focus on inequalities.

2.2.1 Green Transition

As noted in D2.1 (chapter 7), both the academic discourse and the policies around the just transition are mainly related to climate action and green transition. Within the EU, the discourse of sustainability and sustainable development is framed around the European Green Deal (EGD) which focuses on green economic growth, low-carbon infrastructures and technological solutions (Hereu-Morales et al. 2023; Eckert and Kovalevska 2021; Bauhardt 2022; Mastini et al. 2021; Dunlap and Laratte 2022). The EGD is deemed a transformative masterplan in terms of public planning for sustainability, clearly oriented towards achieving the climate objectives of the United Nations (UN) Paris Agreement and towards embedding the sustainability vision of the 2030 Agenda. However, it has been argued that it may not be fit to fulfil its purpose, as the overall focus on economic growth does not acknowledge sustainable development goals that are not directly connected with the economy (Hereu-Morales et al. 2023: 11)[2]. In opposition to a green growth paradigm, the concept of *degrowth* - focusing on minimising human impact on the environment and decreasing consumption to ensure a just transition that also promotes equality - has emerged (Eckert and Kovalevska, 2021; O'Brien, 2022). This will be further explored in Chapter 3.

Furthermore, the green transition should also be considered from the point of view of spatial inequalities, particularly in terms of sustainable development. As noted in various studies, investments in renewable energy and energy efficiency tend to be concentrated in wealthier regions with stronger financial resources and more developed regulatory frameworks. This creates a '**green divide**', which can be illustrated, for example, by how poorer regions are struggling to adopt sustainable practices away from traditional industries, thus experiencing a negative impact of the green transition (Gambhir et al., 2018; McDowall et al., 2023). In other words, spatial justice within the EU context thus requires more scrutiny, with a focus

[2] It should also be noted that the EGD is currently being re-shaped with Clean Industrial Deal, with even more focus on growth related policies. For more details see A 'Clean Industrial Deal' that works for people and the planet

on socio-economic disparities among European cities and regions (Walsh et al., 2021) and existing power asymmetries (Johnson et al., 2020).

Finally, it should be noted that the overall debate on the green transition – especially in the light of justice considerations – tends to focus on the labour market, skills and the future of employment. Studies anticipate a limited impact of the green transition on employment rates overall (Malerba and Wiebe, 2021; Vandeplas et al., 2022), with shifts 'likely to occur between sectors, firms, occupations and regions' (Vandeplas et al., 2022: 5). As a result, the positive and negative repercussions and implications of the green transition on the population may be unevenly and inequitably distributed. If not managed well, the transition to a low-carbon economy may result in job losses and cause hardship, particularly to the communities that currently rely on carbon-intensive activities (Atteridge and Strambo, 2020) such as, for example, the Polish coal basin in the region of Upper Silesia (Jonek-Kowalska, 2015). Furthermore, a low-carbon economy requires workers with the right skills, and the insufficient availability of skilled workers is currently recognised as a bottleneck for the transition (Keese and Macrolin, 2023).

In this context, there is a strong emphasis on the shift towards 'green jobs' and 'green skills'. While the definition of 'green jobs' is not universally agreed, the International Labour Organisation (ILO), defines them as those 'jobs that help to protect and restore ecosystems and biodiversity; reduce energy, materials, and water consumption through high-efficiency and avoidance strategies; de-carbonise the economy; and minimise or altogether avoid generation of all forms of waste and pollution.' (ILO-UNEP, 2008, 35-36). In relation to green skills, these can be broadly defined as 'the knowledge, abilities, values and attitudes needed to live, work and act in economies and societies seeking to reduce the impact of human activity on the environment' (Cedefop, 2022:1). While the need to provide adequate measures to ensure a just transition to green jobs receive ample attention of policymakers, it should be notes that this focus hinders other inequalities, for example related to groups missing from the green economy initiatives (Foundation ONCE and ILO GBDN, 2023). This can be further related to structural vulnerabilities that are likely to occur at the local community level during the restructuring of the local labour markets. It has been noted, for example that while job losses are more likely to affect men in specific communities, significant burdens will also be transferred to women in these locations (Haney and Shkaratan, 2003; Olivier, 2012). These 'hidden groups' should thus be included in the analyses even if they are not directly impacted by unemployment, so that a more holistic, placed-based approach is adopted to gain a full understanding of how structural inequalities are affected by policies and actions undertaken as part of the green transition.

2.2.2 Digital Transition

As noted earlier, there is an overall *assumption* that digitalisation will support the green transition. However, the societal impacts of the digitalisation should not be underestimated, especially in the light of inequalities and structurally vulnerable groups which are not equally benefiting from the digital transformation. Therefore, attention should be given to the way in which technology is supposed to support the consumer end of environmental sustainability solutions, such as smart meters, smart transport, and the necessity to use phone-enabled apps to fully engage with the new developments.

This issue is specifically related to the concept of 'digital divide' , defined as a gap between those privileged and advantaged by the new technologies and those who lack access, skills, resources or abilities to seize their full potential (Calzada et al., 2021; Caragliu and Del Bo, 2023; Kitchin et al., 2019; Rogers, 2001; Van Deursen and Van Dijk, 2019). In terms of gender, women generally tend to have lower ratios of digital skills than men (Eurofound, 2023a). Older people, people with disabilities, low-income people, ethnic minorities and indigenous communities are some of the other vulnerable groups that are most likely to be left behind by the advance of the digital technologies and the new digital economy (Nijman and Wei, 2020; Tsatsou, 2022).

Ageism and ableism are reproduced in most digital experiences and become important barriers for older people and people with disabilities (Kolotouchkina et al., 2022; Kolotouchkina et al., 2023; Nahafi et al., 2022). In terms of age, the literature utilises the concept of the 'grey digital divide' (Mubarak and Suomi, 2022) to describe difficulties with digital systems experienced by senior citizens. However, adopting an intersectional perspective is crucial here as certain groups of older adults (e.g., financially secure retirees who used computers in their working lives) can be more privileged in terms of accessing and using digital tools (Fang et al, 2018). For people with disabilities, the most critical digital barriers are related to the low effectiveness of accessibility tools and assistive devices, personal safety and security online, as well as the cost of adaptive technologies (Lussier-Desrochers et al., 2017; Tsatsou, 2021). These divisions can have important implications for the twin transition, as digital skills will be required if technology is to assist the green transition.

Finally, the **digital divide** can also be analysed in relation to spatial inequality, with important differences among European states, reflecting the overall historical patterns of development in Europe (van Kessell et al., 2022; Lucendo-Monedero et al., 2019). These issues, alongside the urban/rural divide (particularly in relation to the availability of broadband) will need to be further examined.

2.3. Four FITTER Sectors: transition, challenges and inequalities

2.3.1. Energy

The energy sector is at the core of the twin transition, as it is crucial in terms of the reduction of Green House Gas (GHG) emissions and is also needed for the functioning of all the other sectors analysed in the FITTER project, most notably the built environment and transport. As the EU aims to become carbon neutral by 2050, greenhouse gas emissions need to be reduced, and renewable energy is at the forefront of this change (Eurofound, 2023). It should be emphasised that this shift is occurring in the context of the war in Ukraine (Shiffer, 2022) which drastically disrupted energy systems in Europe (Kiltkou, 2022) and which resulted in the necessity for a complex restructuring of energy systems, markets and infrastructures (Kuzemko, 2022), while also leading to high volatility of the energy markets and a sharp increase in energy prices. On the other, the crisis also exposed the need for the EU to be less dependent on natural gas which resulted in an acceleration in the investment and development of climate neutral technologies (Delbeke et al., 2022) as reflected in the REPowerEU Plan, which was developed as a response to this crisis (EIB, 2023). -

In terms of decision-making powers, representation and recognition of diverse social groups as active energy producers, prosumers and decision makers should be accounted for within a just transition framework (Kashour, 2023). However, there are several social groups who are not represented in energy policy and decision making, neither in the private sector nor in the creation of energy technologies. For example, women's decision-making power in this sector, on a global scale, is rather limited (Clancy and Roehr, 2003; Feenstra and Clancy, 2020; Petrova and Simcock, 2021) due to structural inequalities, gender stereotypes, unequal financial resources and the gender pay gap, or the unequal distribution of unpaid care work (DESTATIS, 2023a; DESTATIS, 2023b). It has been estimated that, despite constituting 39% of the global workforce, women only account for 16% of those employed in traditional energy sector with a gender pay gap at the level of 20% (IEA, 2019). While the share of women's employment tends to be higher in the renewables industry (Czako, 2020), they still face barriers in entering both conventional and renewable energy sectors, which can be linked with the lower uptake of STEM subjects among girls at secondary and third level education, as well as retention difficulties due to the masculine working culture (Czako, 2020; ILO, 2024). Women in this sector also face barriers in promotion, advancement and leadership (BCG and WoNY, 2018). There are also other structural inequalities in this context, as people with a migration background, as well as those coming from households with lower level of academic education, are less likely to be represented in leadership positions of renewable energy companies (e.g. Genkova et al., 2022).

With regards to the consumers' side of the energy sector, one of the key concepts related to energy and inequalities is **energy poverty,** defined as 'a situation in which households, or individuals, cannot attain and/or use energy services required for good health, wellbeing, and the ability to fully participate in society' (Simcock et al, 2021:2), for example, in terms of inability to sufficiently heat a home or to afford the cost of other necessary household energy services (Redemaekers et al., 2016). Energy poverty can be shaped by existing inequalities: for instance, lone parents are more likely to anticipate difficulties in paying bills compared to the general population (EIGE, 2023a). While green energy can be essential in addressing the immediate and long-term challenges of energy poverty (Carfora and Scandurra, 2024), several groups have been identified as vulnerable to energy poverty in the context of the green transition the Global North. These include low-income households, people from ethnic minorities and indigenous people, immigrants and those with low proficiency in the local language, older people, disabled persons, women, young people and full-time students, single-parent families, socially isolated people and those living alone, as well as large households and multi occupancy households (Jessel et al., 2019; Middlemiss, 2022). Intersecting inequalities need to be considered: for example, single-parent households are likely to be led by women, often from ethnic minority background, and in low incomes (Middlemiss, 2022). In addition, research has shown that poorer households tend to spend a higher share of income on energy compared to richer households, especially in the context of rising energy costs, which can amplify energy poverty (ILO-OECD, 2022) as is usually the case for all utilities (Fankhauser and Tepic, 2007).

2.3.2. Construction and Built Environment

The construction sector comprises an important part of the EU economy. It accounts for around 6% of the EU Gross Domestic Product (GDP) and employs around 13.5 million people – 6.6% of total employment in the EU27 in 2021. The sector is also important for achieving green transition goals, as it has been estimated that buildings and construction contribute 37% of global energy-related carbon emissions (IHRB, 2024). The EU strategy 'A Renovation Wave for Europe: Greening our Buildings, Creating Jobs, Improving Lives' includes plans to improve the worst performing buildings, address energy poverty and decarbonise heating and cooling systems.

In relation to employment in the sector, it is estimated that construction is set to drive most of the net employment growth resulting from the implementation of the FIT for 55 package, with a concentration in low-mid and mid-income jobs (Eurofound, 2023c). There are certain skills needed for the successful transition in the sector, such as specialist building renovations and heating and cooling expertise, as well as the increased involvement of the 4.0 industry (Eurofound, 2023c, Turner et al., 2021). The main equality issues discussed in the context of the transition in the sector are gender and age, as women have been traditionally underrepresented in the sector, while young people are no longer attracted to it (Mella and Werna, 2023). Migrant workers are also referred to in terms of vulnerabilities, due to 'routine' breaches of employment legislation in the sector (Davies, 2021).

On the receiving end of the sector, the necessity to renovate is linked to the concept of 'decent housing', with several studies pointing out to the issue of inadequate and unfordable housing affecting many across Europe. As noted by Eurofound (2023) unaffordability of housing can lead to housing insecurity and affects people's health and wellbeing, with the cost of living further exacerbating inequalities. In general terms, data shows that only a quarter of the adult population in Europe lives in buildings that have been renovated within the last five years (Eurostat, 2024).

The issue of inadequate housing and its relationships with energy poverty should be further explored. As living in low-quality buildings is also related with energy poverty, renovation of existing housing stock is among the priorities of a just transition, although inequalities in this area are likely to persist (Friends of Earth Europe, 2024). Ownership status is an important factor here, as renovation policies are focused on homeowners and thus are less likely to have a positive impact on tenants in private rented accommodation or in social housing (Economidou and Bertoldi, 2015). This intersects with socio-economic status, as lower-income households are less likely to be homeowners, and also have limited access to credit, which is often needed to invest in retrofitting (ILO, 2022). Some authors also refer to the processes linked to large-scale renovations targets as to 'low-carbon gentrification' (Bouzarovski et al., 2018), which leads to the involuntary displacement of historically marginalised groups, such as working class, low-income, racialised minorities, youth and older residents, as well as the loss of social networks, social capital and community cohesion, eroded sense of place and identity, socio-cultural exclusion, increased criminal activities, and barriers in accessing healthcare services (Anguelovski et al, 2021). This is also related to 'renoviction' a term describing evictions based on the grounds of renovation – usually discussed in the context of for-profit renovation practices (Polanska and Richard, 2021).

2.3.3. Transport

The transportation sector is critical to European business and the supply chain, accounting for around 5% of the EU GDP and employing more than 10 million people. Transport also accounts for almost a quarter of Europe's GHG emissions, so low-emission mobility is essential to achieve a broader goal of a shift towards low-carbon economy (European Commission, 2016). In relation to legislation and a move to sustainable transport, the EU established the Fit for 55 package, which sets up targets for emission reduction in the sector, and introduces incentives for the uptake of low-carbon and renewable fuels.

Transitions in transport and mobility affect people in multiple ways, as shifts are having an impact on those directly employed in the sector, those who are involved in the manufacturing of vehicles (e.g. cars) and those who are end users of the different modes of transport.

In terms of employment, challenges include a move towards sustainable fuels with the goal to reduce GHG emissions. Within the EU, guidance for reduction in GHG emissions from transport are provided in the Sustainable and Smart Mobility Strategy (SSMS) adopted in 2020 under the EGD. The current shift towards electric vehicles will result in substantial job losses and the need for re-skilling and upskilling (Broughton et al., 2024). Furthermore, employment in the sector presents significant inequalities in terms of gender and age, as it tends to employ older male workers (Broughton et al., 2024). Improving gender balance is arguably a significant challenge, as women constitute around 22% of the total transport workforce and suffer barriers such as misogynistic attitudes, male dominated hierarchies and gender pay gap, among others (Woodcock and Rosenau, 2023). In addition, decision-making process in the design of transport and of vehicles is male dominated, therefore resulting in gender inequalities at the user level (Woodcock, 2023).

Access to adequate transport is essential as it can help to reduce inequality and enhance socio-economic mobility, and, in this regard, **transport poverty** needs to be considered when analysing structural vulnerabilities in relation to transformations in the sector. Transport poverty is characterised by transport-related social exclusion (Verhorst et al., 2023) and can be defined as 'the inability to attain a socially- and materially necessitated level of transport services' due to affordability, mobility or access (Simcock et al., 2021). The cost of transportation, for example, can strongly influence individual choices. When these costs are high compared to available income (all other characteristics such as accessibility, adequacy and availability being constant), individuals' mobility and access to services (e.g. healthcare, education) and opportunities (e.g. employment, cultural events) might be limited (Kuttler and Moraglio, 2021a). Transport availability and accessibility is also a factor reinforcing poverty and social exclusion (Lucas et al., 2016) and is a significant barrier for disabled persons (Kiss, 2022). Furthermore, time spent on traveling is also considered as a factor impacting inequalities, with 'time poverty' used as a term in this regard (Kiss, 2022). Combined with the availability of transport, time spent on travelling is also related to as a dimension of transport poverty (Furszyfer Del Rio et al., 2023). In this context, gender has been identified as an important inequality ground. Women are the main users of the public transport but are more likely to experience time poverty as they tend to take shorter and multiple trips related to household and care duties, thus spending more time on travel; while men engage with linear trips between work and home, which can be linked with the traditional division of labour (ILO, 2024).

Structural inequalities are clearly linked with both transport and energy poverty. Studies have found that these often overlap, with groups identified as at risk of energy poverty, including people with low income, those with pre-existing health conditions and disabilities, lone parent families, older citizens, women, those with ethnic minority background, those in urban settings and more particularly suburban households more likely to experience transport poverty (Koukoufikis and Uihlein, 2022). This is also often referred to as 'double energy vulnerability' with certain groups more likely to be affected by it than others, for example those from income households and ethnic minorities (Furszyfer Del Rio et al., 2023).

However, affordability, mobility or access are not the only factors behind the ways in which people access transport services. Research on active transport shows that the benefits of walking and cycling are not distributed evenly across the populations, as socio-economic and cultural factors play an important role in this context (e.g., underrepresentation of disabled people in active transport journeys, men cycling more than women) and that intersectional lenses need to be adopted to analyse these interconnections (Yuan et al., 2023).

Finally, the push towards electric vehicles for private usage also requires more scrutiny in relation to potential benefits and costs, and its impact on inequalities. The price of electric cars also remains high and may be unaffordable for certain segments of society. This is despite subsidies available, as these are often conditional upon an upfront payment which requires significant capital investment before the subsidy is applied (Dwarkasing, 2023). Studies have also demonstrated that men are more likely to be an EV owner as they have higher purchasing power and thus can afford the upfront cost of an electric car (S&P Global Mobility, 2022). Some authors also suggest additional inequality grounds as important in this context, as those most likely to be EV owner are men in urban areas, with higher educational attainment, higher income, full-time employment, often in a household which already owns more than one car (EIGE, 2023b). This issue would require further study, particularly in the context of future penalization of fossil fuel ran vehicles, as this can potentially affect groups already at risk of transport poverty and result in their further marginalisation.

2.3.4. Agriculture and food sector

Agriculture is a strategically important sector of the economy and holds a substantial social and cultural significance. Its activities also have significant implications for nature, air, water and climate. It has been estimated that agriculture and other land use contributes to approximately 23 percent of the global emissions (IPCC, 2020) while the EU food systems generate approximately 30% of the GHG (Crippa et al, 2021), the majority originating from animal-based foods (Leip et al., 2015). Changes required to the agriculture and food sector are noted to require a systemic change, with a major shift away from the 'productivist' development paradigm defined as farming systems aiming at maximizing yields and profits (Helfenstein et al., 2024) and towards more environmentally friendly farming. At the EU level, policy developments in this regard include: The Farm to Fork Strategy, the EU biodiversity strategy for 2030, the Common Agricultural Policy (CAP) 2023-2027 and European climate law (EEA, 2022). It should also be emphasised that while farmers and farming practices are at the core of these shifts, changes are needed across entire food systems and should include such actors as supermarkets and consumers (Hebnick et al., 2021; Kuhmonene and Silataoja, 2022) in order to challenge cultural models of eating well (Kaljonen et al., 2020).

This has been identified as a sector particularly vulnerable to the impacts of the green transition, due to its carbon emission and high path-dependency which can be linked with being tied to markets and contracts, capital investment or skills availability (Knickel et al., 2017; Sutherland et al., 2012). The need for a dietary transition is also identified as a potential threat to farmers (Kuhmonen and Siltaoja, 2022), most likely to affect those involved in meat production. Furthermore, it must be noted that the agriculture sector is particularly dependent on nature and environment, and thus itself depends on climate sustainability (McCabe, 2019). Across Europe, farmers experience growing competitive pressures and face challenges related to labour shortages and increasingly rely on migrant workforce, in some cases of informal nature (Cohen et al., 2022).

With regards to inequalities, most of the concerns relate to the farmers themselves as they are central to the transition yet often constrained by economic and political circumstances (Williams et al., 2023) while forced to bear the costs of this change (Helfenstein et al., 2024). With regards to finances, while the average income per farm worker in the EU has been increasing, certain income inequalities persist, with significant differences between the bottom and the top 10% (EC, 2024). The distribution of CAP supports is also argued to be unequal, as it has been estimated that 20% of farmers benefit from 80% of CAP supports (Garcia-Bernado et al., 2021) although the new CAP recognises the need for a fairer distribution (Viegas et al., 2023) with several studies showing a positive effect (Hanson, 2021). Regional divides are also identified as a crucial factor contributing to inequalities, as the East-North/West divide is persisting, with the general pattern of higher profits recorded by larger farms typically located in Germany and France (EC, 2024).

Gender imbalances are also prevailing, despite the recent emphasis on the role of women in the sector. For example, with regards to farm management, women in the EU on average represent 30% of farm managers (Fanelli, 2022). Farms are also more likely to be led by men, with farm mangers also located in the older age brackets, with the majority over 55 years old (Eurostat, 2024c). In the European context, women in rural areas are also more likely to be unemployed than men, as well as over-represented in informal and vulnerable employment. (Casares Guillen, 2021). Concerns are also raised regarding income inequalities in the sector, as the poorer households are more likely to foot the transition bill, with small, family-owned farms facing new set of environmental restrictions (Leitheiser et al., 2022; Van der Ploeg, 2020). Gender pay gaps are also of concern, as farms run by women have income of more than one third lower than those run by men (EC, 2024). Women are also more likely to be engaged in informal labour (e.g. seasonal planting and harvesting), often unregulated (ILO and FAO, 2021). Furthermore, women are underrepresented in the EU agriculture policymaking (Guillot, 2022), and policies are often designed with male farmers in mind, thus failing to reflect women farmers knowledge and perspective (ILO, 2024).

However, recent studies show that women and younger farmers are more involved in sustainable agriculture, with younger female farmers more oriented towards environmentally sustainable practices compared to younger male farmers (Unay-Gailhard and Bojnec, 2021: 72) although some caution is needed with regards to conclusions drawn from these findings. International evidence suggests that gender continues to be an issue even in sustainable farming. Within the EU, women represented only 26.9% of fully organic farm managers by 2020, and this proportion was lower than the share of women among managers of all farms (Eurostat, 2024). This highlights the importance of scrutiny in examining the actual farm family orientations and practices to transform gender relations.

At the receiving end of the food chain, **food justice** is recognized as an important issue. Food justice can be defined through the broader context of social justice, emphasising that sustainable and equitable food is a human right (Murray et al., 2023). Yet, climate change can pose a threat to food production and is likely to increase food prices (Kotemaki, 2019). Dietary changes are among three main levers of the EGD, with a shift toward diets containing lower quantities of animal-based products at the core (Guymomard et al., 2023), although socio-economic inequalities should be taken to account in this context, as in high income countries, those from lower income groups are more reliant on consumption of red processed meat (Clonan et al., 2016). Food justice can be further linked with the relationship between unhealthy diet and socio-economic status (Dorman and Drewnowski, 2008), with research also showing a link to the time and skills required to prepare food, with those from lower socio-economic backgrounds being negatively affected (Wenworth and Chapman, 2024).

3. The Twin Transition: perspectives on sustainability and the role of technology and social innovation

As outlined in D2.1, global temperatures for the past five years were the highest on record and the negative impacts of global warming are undisputable. Work completed in FITTER, subsequent to D2.1, has demonstrated the needed for in-depth understanding of the twin transition - with a particular focus on understanding the meaning of sustainability, the role of technology in eco-social policy development, and potential avenues of social innovation. The concept of social innovation is explored through examining divergent understandings of the economy, labour market and social supports.

3.1. Background

The focus of D2.1 was an investigation of social relations and approaches to policy making in line with the projects aims of contributing to existing research on the **origins, dynamics and determinants of inequalities** and enabling **anticipatory governance** to support a fair and inclusive twin transition across Europe.

3.1.1. The origins, dynamics and determinants of inequalities

In D2.1, we examined the structure of the social contract as a shaping mechanism for the division of labour and participation in the public sphere (origins of inequalities). At the centre of the Western construct of the social contract is an understanding of the individual as autonomous, unencumbered, and someone who 'opts in' to the social contract of their own volition, thereby agreeing to fulfil certain responsibilities, and able to avail of their rights. The liberal social contract has been shown to be based on both a sexual and racial contract which has led to the systematic exclusion of women, people of colour and indigenous peoples in social, political and economic spheres (Pateman, 1988; Mills 1997; Squires, 1999; Bhambra, 2021). It has also led to the allocation of little to no value to the environmental and social costs of extractive industries, and to the activities of the reproductive sphere e.g. care work, the provision of social supports, education.

We foregrounded the importance of understanding how power works by outlining the 'power cube' developed by Gaventa (2006, 2021) and providing additional analysis (Allen, 2001; Bradley 2020; Zerilli 2006). To understand what level of transformative action a policy can achieve it is important to understand the interrelationships of spaces, levels and forms of power, and the operation of different modalities of power (dynamics of inequalities).

We also examined the long-term consequences of histories of colonialism and the embeddedness of capitalism as an economic system (origins and determinants of inequalities). Drawing on the wider literature we highlighted the dimensions to inequalities within the climate-environment-social nexus (Galgóczi and Akgüç, 2021) and the triple injustice of a 'business as usual' (BAU) economic growth-focused approach to policymaking (Mandelli, 2022: 340).

More detail on these key concepts can be found in the tables in **Appendix A**.

3.1.2. Enabling anticipatory governance

In D2.1, we foregrounded the UN/ILO Guidelines for Just Transition and the importance of FITTER governance recommendations aligning with them. We provided an overview of the conceptual development of environmental justice, climate justice, energy justice and spatial justice and we identified the importance of key dimensions of justice – justice as recognition, justice as redistribution, justice as representation. We also identified restorative justice as an important element of ensuring fair burden sharing (Fraser, 1995; Heffron and McCauley 2017). Governance structures need to allocate time and resources to enabling these processes.

We reviewed the literature on eco-social policy development and focused on the definition of an output focused eco-social policy developed by Mandelli (2022). An output focused eco-social policy has two defining features - first that it **explicitly** pursues both environmental and social objectives, and second that it does so in an **integrated** way. We outlined the six quality criteria for intersectional policy design and evaluation developed by Lombardo and Agustín (2012). We outlined the criteria for drawing up a list of functionings and indicators in a research study developed by Robeyns (2003). We highlighted the argument provided by Wolff and De-Shalit (2007) that while the functionings people can achieve is important, it is equally as important that those functionings are secure i.e. they can sustain their current level of functionings. Thus, the risk and vulnerability of losing key functionings also needs to be accounted for i.e. the environmental costs of current activities for future generations. This is particularly relevant to forecasting scenarios in the context of the just transition wherein ecological protections are required to ensure current and future peoples can achieve basic functionings (Holland, 2008).

The governance of the just transition will need to account for and integrate justice frameworks, conduct intersectional eco-social policy design, and select evaluative functionings or indicators in line with a robust methodology.

More detail on these key concepts can be found in the tables in **Appendix A**.

3.2. Understanding the twin transition

Historically environmental justice movements have focused on analysis and campaigns to ensure justice at a local or regional level, and the climate justice movement has focused on cross-border global issues. However, there is an increasing overlap between contemporary environmental and climate justice concerns, and they have fused in many ways. A key part of this has been a shift to thinking of **adaptation as transformative** in line with a revised eco-social contract. There is recognition that justice movements need to be both local *and* international in order to build social systems that are 'based on the protection of sustainable socio-ecological systems, governmental accountability to all those impacted by climate change, and a notion of security built on basic human needs' (Schlosberg and Collins, 2014: 370). The focus is on building local capacities, and development out of poverty and injustice, as the **basis for adaptation**.

The EU has committed to achieving net-zero emissions by 2050; maintaining, but also decoupling, economic growth from resource use; and delivering a just transition which leaves no-one behind. These are ambitious goals and current targets are not anticipated to be reached (Climate Action Tracker, 2024).

As discussed later in this section, policy responses in EU countries focus on ecological modernisation and degrees of social investment, albeit without sufficient acknowledgement of the climate emergency as an emergency, without the application of substantive justice principles, and without the substantive reconceptualisation of the welfare state that a reworking of the social contract in the direction of a eco-social contract would entail. In part this is due to how policy-making currently operates as a top-down, largely elite-driven, often single issue, process. It is also partly attributable to different understandings of **sustainability**; diverse understandings of the role **technology** could play in service of environmental targets; and distinct understandings of the role of **social innovation** in addressing climate breakdown.

3.2.1 Sustainability

Many applications of the concept of sustainability can be aligned with what Naess (1973) terms 'shallow ecology' which is defined as combatting pollution and resource depletion to enable the wellbeing of people in the Global North.

In response to this limited focus, Naess developed the concept of 'deep ecology' highlighting key principles:

1. Humans exist within a relational, total field image of the environment – all organisms including humans can be understood as knots in a net of intrinsic relations
2. Biospherical egalitarianism in principle – ways and forms of life have an equal right to live and thrive although some level of destruction and killing is unavoidable
3. Diversity and symbiosis – we should facilitate an ability to coexist in complex relationships; support diverse ways of life among humans and other species; live and let live
4. Anti-class posture – relations of oppression harm both oppressors and the oppressed and wide classless diversity should be enabled
5. Fight against pollution and resource depletion
6. Complexity, not complication - organisms, ways of life, and interactions in the biosphere generally interact in complex and integrated ways. We need integrated actions and complex economies which integrate a variety of means of living.
7. Local autonomy and decentralisation – there is a need to strengthen local self-government and self sufficiency (Naess, 1973: 95-98).

Val Plumwood (1991) argues that deep ecologists understand the core driver of ecological destruction to be human separateness from nature (atomism). The solution is therefore proposed as the identification of self with nature. However, the principles of deep ecology have been criticised for replacing one individual – the autonomous individual of Western thought – with another super-individual – the ecosystem (Slicer, 1995). Plumwood classifies three, often overlapping, variants of the conceptualisation of the 'identification of self with nature' by deep ecologists:

- The indistinguishability account – argues we are all connected and also ultimately indistinguishable entities – humans are but one strand in a biotic net.
- The expanded self – therein identification does not mean 'identity' but rather 'identity with empathy' such that through the expansion of self we take the interests of nature as our own interests. The expanded self operates in the interests of nature because they are in line with their self-interest.
- The transcended or transpersonal self – advises an overcoming of self, striving for impartial identification with all particulars and the whole cosmos, discarding our own interests and attachments to particular concerns (Plumwood, 1991: 12-16)

Plumwood argues that deep ecologists in fact misunderstand the problem as 'atomism' and therefore propose an incorrect solution – identification (variously presented along a continuum, from the expansion of self-interest to wholesale self-sacrifice). Against this position she argues that the driver of ecological destruction is in fact the construct of 'dualism' and particularly the human/nature dualism (1991: 17-18).

Plumwood presents three features of the human/nature dualism that need to be analysed and addressed if we are to address the drivers of environmental destruction.

- Concept of the human: within the human/nature dualism the ideal of the human is one who is rational and highly analytical. The mental functionings of the human are highly valued and there is an accompanying devaluation and alienation from biological functionings i.e. that which we hold in common with animals/nature. That which is understood as the human essence (mental functionings) is positioned as superior to nature and the body (biological functionings). With this understanding of the human, in order to be authentically human we must maximise our control over the natural sphere – both within ourselves and across the wider environment. Ideally we should transcend the material sphere.
- Concept of nature: within the human/nature dualism nature is understood as discontinuous with the human, and also inferior. Nature is constructed as passive and lacking in agency and teleology. The value of nature lies only in its usefulness to humans.
- Concept of self: to uphold the human/nature dualism a particular concept of the self is needed. This self is egoistic and engages in an instrumentalist relationship with other people and the environment. They are invested in maintaining a strong distinction, and maximal distance, between the superior self, and inferior 'Others' and nature (1991: 17-22)

Plumwood argues that in understanding the problem to be 'atomism' rather than 'dualism' deep ecologists fail to address

- the superior/inferior oppositions that dualism constructs;
- our continuity with the natural world but also its distinctiveness from us and its distinct needs;

- the need to challenge the assumption that human nature is primarily and inherently egoistic and the only alternative to this is self-sacrifice; and,
- the fact that human care and connectedness to others and to particular concerns, social ties, and places, is as important a basis for flourishing as any other (Plumwood, 1991)

This is not merely a conceptual or philosophical debate – it has practical impacts with regard to the prioritisation of goals for the green transition, and how a just transition is managed.

By assuming egotistical and self-interested human nature as the dominant form, deep ecologists risk facilitating an anti-humanism that justifies violence against humans in the interests of the rights of nature to thrive. Their account fails to properly differentiate how humans are differently positioned through our histories, societal structures, norms and power relations. By assigning equal value to all living beings, and failing to differentiate how, as humans, we hold different levels of responsibility for ecological destruction, the rights of nature can be pitted against the rights of marginalised peoples. Interventions in nature can only be justified on the basis of preserving the integrity of nature, not in order to meet the needs of humans (Guha, 1989 cited in Tokay, 2024). Such a position disregards the deep and meaningful relationships people have to place, which can be specifically non-instrumentalist, as exemplified by indigenous accounts of connections to land that are as deep and meaningful as their ties to family and friends (Neidjie 1985 cited in Plumwood, 1991).

Project Tiger: An example of a deep ecology approach to conservation
Ramachandra Guha (1989 cited in Tokay, 2024) argues that deep ecologists tend to privilege nature which is free from human interference. This elevates the wilderness over cities and suburbs as sites of sustainable development and environmental protections. It thereby significantly narrows the priorities and scope of conservation strategies. Guha outlines how 'Project Tiger' exemplifies the priorities of deep ecologists. This project was designed to protect the Bengal tiger species by allocating a large amount of land for tigers to roam. The allocation of land entailed the displacement of villagers living in or around the reserves, and resulted in a transfer of resources away from the villagers who relied on natural spaces for their subsistence. Guha argues that by the standards of deep ecology the project was and is a success in that the numbers of tigers increased. However the success of the project was predicated upon the exclusion and displacement of villagers and the transfer of resources they relied upon to the tourist industry and the state. In failing to account for histories, power relations and different degrees of responsibility for ecological destruction the needs of marginalised peoples can be disregarded and in fact positioned as in conflict with the needs of nature (understood as ideally free from human interference) (Guha, 1989 cited in Tokay, 2024: 99-100)

In contrast to the egotistical self that is assumed by deep ecologists, Plumwood (1991) proposes an account of 'self-in-relationship.' She argues that this

> ..is an account that avoids atomism but that enables a recognition of interdependence and relationship without falling into the problems of indistinguishability, that acknowledges both the continuity and difference, and that breaks the culturally posed false dichotomy of egoism and altruism of interests; it bypasses both masculine 'separation' and traditional-feminine 'merger' accounts of self. It can also provide an appropriate foundation for an ethics of connectedness and caring for others (1991: 15)

Instead of privileging nature as free from human interaction, the self-in-relationship account foregrounds the realities and unavoidability of human-nature interaction. It orients us to imagining and constructing 'productive human/nonhuman interaction' (Tokay, 2024: 104). As Ela Tokay (2024) argues we need to foreground the agency of all involved in a situation rather than trying to establish a hierarchy of value (i.e. nature over people, tigers over villagers). Assessing how diverse elements interact over time in complex ways is not easy. However engaging with complexity

> could lead environmentalists or policy makers who are in a position to decide the best course of action (positions of power, the basis of which we are also right to question) to proposals different from the ones that take the displacement of the human inhabitants as the only viable solution (Tokay, 2024: 105).

A standard definition of sustainable development developed by the Brundtland Commission, and which has been adopted within key policy frameworks, proposes that

> ..the ability to make development sustainable to ensure that it meets the needs of the present without compromising the ability of future generations to meet their own needs (1987)

The Brundtland Commission took a largely instrumental and utilitarian approach to sustainable development but there is still an understanding of the importance of achieving environmental sustainability, social sustainability and economic sustainability in tandem and in an integrated manner (Brundtland et al., 1987). This is in line with the contemporary focus on integrated eco-social policy design. Critiques of deep ecology demonstrate the importance of taking a systems wide approach to addressing environmental destruction. By engaging with histories, power relations, agency, roles, and responsibilities, we can consider how diverse actors and elements interact and consider a range of solutions or pathways (Tokay, 2024).

3.2.2 The role of technology

The European Green Deal is focused on the development of green technology, including carbon capture technologies, in order to achieve climate neutrality by 2050 (European Commission, 2021). The main goals of the European Green Deal are:

- Climate Neutrality - Drastic reduction of greenhouse gas emissions for the EU
- Circular Economy - New economic model where products are reused, repaired and recycled, reducing waste and conserving resources

- Clean Industry – Develop cleaner, more sustainable and energy-efficient industries which thrive in the EU and global markets
- Healthier Environment - Plan to restore nature and work towards zero pollution to ensure a healthy environment for future generations
- More Sustainable Farming - Greener farming practices to protect the environment while providing healthy and affordable food
- Climate Justice and Fairness - Plan to make the transition fair and inclusive to help people most affected by the transition and leave no one behind

Net zero refers to a state in which greenhouse gas (GHG) emissions are reduced and those emitted are balanced out by equal removal of GHG from the atmosphere through carbon capture, utilization and sequestration (CCUS). In order for it to be effective net zero must be permanent i.e. removed GHG must not return to the atmosphere through measures such as deforestation or improper carbon storage (NetZeroClimate, ND).

There is debate and mixed evidence as to the efficacy of CCUS methods. Projects are reliant on still early-stage technology. Some deliver environmental and economic benefits, but others – such as enhanced oil recovery - worsen climate breakdown. Scientists and activists have warned that it is important that governments do not rely on the potential of carbon-capture technologies as an alternative to the urgent task of phasing out fossil fuel use completely (Sick, 2024; Khoury, 2023).

Much of the 'tech' in 'green-tech' is reliant on the extraction of raw materials from the Global South and this is a key concern of **climate justice** and **energy justice.**

As Ankersmit and Parstiti (2020) demonstrate there are significant shortcomings in the EU's approach to promoting the sustainable extraction of raw materials as part of Free Trade Agreements (FTAs). These shortcomings include:

1. the exclusion of interests other than that of EU industry in dedicated raw materials chapters;
2. the almost exclusive focus on trade *liberalization* commitments over trade *regulation* in these chapters;
3. the separation of economic interests from social and environmental interests in raw materials extraction by moving the latter to more general, secondary, weak and widely criticised sustainable development chapters, instead of an integrated and dedicated approach to transnational economic activity in the mining sector.

Ankersmit and Parstiti propose that 'sustainable extraction of raw materials should be a precondition to trade and investment in those materials and not an afterthought or based on voluntary and individual efforts by the trading countries concluding such agreements' (2020: 9). They propose elements to ensure:

1. that sustainable extraction of raw materials is a precondition to trade and investment in those materials;
2. that the interests of all stakeholders affected by trade and investment in the extraction of raw materials are reflected in the language of the provisions, in particular the interests of local communities, indigenous peoples, the environment, and labour;

3. that commitments by the parties are tied to, complement, and foster international efforts to make extraction of raw materials more sustainable;
4. that commitments are a floor and never a ceiling for the regulation of the extraction of raw materials;
5. that monitoring and enforcement of the provisions move away from a strictly economic rationale, towards a rationale based on compliance.

Furthermore, as noted in chapter 2 there is mixed evidence as to the degree to which technological advancements, or the digital transition, actually enable the green transition. Although dominant policy frameworks, including the sustainable development goals, Agenda 2030 and the European Green Deal accept growth as inevitable, and assign technology a foundational role in achieving green growth, a sole focus on technological development risks neglecting the role of equally important initiatives such as social innovation (Sætra, 2023a: 16; Mandelli, 2022; Murphy, 2024).

Four stances can be identified with regard to the role of technology in enabling a green transition – techno-optimism, techno-pessimism, techno-realism and techno-colonialism.

As Danaher (2022: 54) outlines, techno-optimism can be defined as a stance that holds that

> technology, defined here in largely material and instrumentalist terms, plays a key role in ensuring that the good does or will prevail over the bad … [where] 'technology' has to be understood at a reasonably high level of abstraction — not just particular gadgets but the totality of processes of technological production.

Techno-optimism can be compatible with both agency-based theories of technological development (where such development remains under human and social control) and techno-determinism (which refers to the idea that at least some processes of technological formation are beyond human control). However, as Danaher notes,

> there may be a tendency for techno-optimists to lean into determinism. Given the messy contingencies of human choice and action, it might be difficult to anchor optimism in any view that gives humans control over the process of technological development, particularly if we have a cynical view of human nature (Danaher, 2022: 10)

Techno-pessimism is the belief that the totality of process of technological production does not lead to an improvement in the human condition.

Techno-realism recognises the embeddedness of technology in society and the political nature of technology. There is an emphasis on the requirement for the social control of technology such that we enable sustainable, democratic and inclusive futures (Sætra, 2023b: 262; Bennahum et al 1998).

Madianou (2019: 2) develops the concept of techno colonialism to refer to the ways in which the 'convergence of digital developments with humanitarian structures and market forces reinvigorates and reshapes colonial relationships of dependency.' Brevini (2023: 28) argues that these dynamics also 'characterize the global supply chains of Artificial Intelligence, as the extractive nature of techno-colonialism resides in the minerals that need to be mined to make the hardware for AI applications [...] thus compromising several SDGs (13, 15, 12 to list a few).'

A significant critique of 'future' techno-optimism with regard to addressing climate breakdown - the belief that future technologies will 'fix' carbon emissions and environmental harms without changes in lifestyle - is that it is reliant on predictions about future technological developments which are inherently unknowable or cannot be predicted with an adequate degree of certainty (Keary, 2016 cited in Danaher, 2022).

Drawing on Lisa Bortolotti's (2018) account of an agency-based theory of optimism, Danaher (2022: 54) proposes a concept of a '*contingent, techno-social* form of optimism' arguing that a techno-optimist must situate technology within and relative to the human collective, work to identify goals which will enhance the collective good and use technology in a way to achieve those goals.

Danaher's modest, agency-based version of techno-optimism is close to the account of techno-realism developed by Sætra (2023b). Sætra argues that achieving 'sustainable development – broadly understood – might indeed require technology, but it cannot be based on a blind faith in technology-based growth' (2023b:262). They propose that we cannot do without technology but we need to ensure that we identify and develop the right kinds of technology and we need to analyse technological development as part of broader socio-technical systems rather than as isolated instruments abstracted from values and ideology.

Sætra (2022 cited in Sætra 2023a) stresses that an analysis of the impacts of technological development must incorporate the micro-, meso-, and macro-levels.

At the individual or micro-level, technology may improve life for some people e.g. speech output for blind and visually impaired people. However, the increased use of technology can also further reinforce existing dynamics of exclusion at the individual level. As van Dijk (2020: 13) argues

> 1. Inequalities in society produce an unequal distribution of resources. 2. An unequal distribution of resources causes unequal access to digital technologies. 3. Unequal access to digital technologies also depends on the characteristics of these technologies. 4. Unequal access to digital technologies brings about unequal outcomes of participation in society. 5. Unequal participation in society reinforces inequalities and unequal distributions of resources.

At the group (meso) level, technology may improve the life of specific groups of people e.g. the ability to identify and join a support group online. At the group level though technology may also enable discrimination e.g. bias identified in automated hiring systems and in the algorithms uses for recidivism prediction and social welfare management (Dastin, 2018; Lambrecht and Tucker, 2019; Dencik, 2022; Rachovitsa and Johann, 2022; van Dijck, 2022)

At the social (macro) level, technology can be used to develop smart cities and renewable energy production. However, technological development also accounts for significant levels of resource extraction and energy use on a global scale. As Brevini (2023: 29) notes greenhouse gas emissions from the Information and Communication Industry (ICT) could grow from roughly 1–1.6% in 2007 to exceed 14% worldwide by 2040, accounting for more than half of the current relative contribution of the whole transportation sector.

The growing literature on the need to better evaluate the degree to which the digital transition does in fact enable (or undermine) the green transition is very much in line with the systems approach to energy justice developed by Heffron and McCauley (2017) and which FITTER has adapted.

Heffron and McCauley place the energy life cycle at the centre of the framework. This includes 1) Extraction; 2) Production; 3) Operation and supply; 4) Consumption; 5) Waste Management. They advocate a three-step analysis of a given issue: 1) Analysis of the issue vis-à-vis distributional, procedural and recognition justice tenets; 2) Place the issue in the broader energy life cycle taking global interdependencies and issues into account; 3) Employ applied principles for guidance on practical action and restorative justice as a condition for action.

Although individual policymakers will not be able to conduct a full analysis of the social and carbon footprint of component parts of technologies, significant awareness raising on the social and environmental costs of resource extraction is needed and achievable in the context of a just transition. Arguably the onus should be on technology companies to enable informed policy making and consumer choices. Brevini (2023: 30-31) suggests several governance mechanisms through which the environmental and social harms of energy and ICT extraction, production and waste disposal can become more transparent.

- Accountability for those who own clouds and data centers
- A government-mandated Green Certification for server farms and centers to achieve zero emissions.
- The disclosure of server farms and centers carbon footprint e.g. a Tech Carbon Footprint Label, which would include information about the raw materials used, the carbon costs involved, and what recycling options are available
- Stronger public awareness about the environmental and social justice implications of adopting a piece of smart technology.
- Transparent calculations of the energy used to produce, transport, assemble, and deliver technology
- Requirements that manufacturers lengthen the lifespan of smart devices and provide spare parts to replace faulty components.
- Educational programs to enhance green tech literacy and raise awareness of the costs of hyperconsumerism, as well as the importance of responsible energy consumption
- Green tech literacy programs could also entail interventions to ban production of products that are too data demanding and deplete too much energy.

3.2.3 The role of social innovation

In line with the account of techno-realism developed by Sætra (2023b) and aligning with the account of a ‘*contingent, techno-social* form of optimism’ developed by Danaher (2022), an assessment of the degree to which the digital transition enables the green transition, and whether it does so in a just manner or not, entails a critical assessment of the origins, determinants and dynamics of inequalities and environmental harm and destruction. FITTER mitigation measures must account for the possibilities for, and barriers to, both digital innovation and social innovation in enabling a just transition.

The likelihood of global communities' ability to enable a transition to a net-zero economy, or achieve a *just* transition to a net-zero economy, - or their inability to achieve either - is significantly shaped by the dominant economic pathway taken. Preferred economic pathways are shaped by political ideology, political commitments/priorities, and understandings of justice. Three frameworks for an economic pathway are discussed in the environmental and social policy literature – green growth, the foundational economy, and doughnut economics. They each have distinct understandings of key concepts such as the economy and the relationship of the labour market to social supports and these understandings significantly shape the mitigation strategies they propose.

The Economy

A **green growth** economic pathway takes the classical economics understanding of the economy as defined by competition and trade in goods and services. Therein 'the economy' is understood as a separate sphere to social relations and the environment. A focus on green growth enabled by technology is popular because it offers the potential of maintaining current lifestyles in the Global North with adjustments, rather than radical transformative actions, required from business and consumers. Within this paradigm economic growth remains the primary goal but decoupling it from environmental harm and destruction is also required. Evidence indicates that a green growth paradigm, driven by technology (but without demand-reduction strategies) keeps us on a collision course with planetary limits (Hickel and Kallis, 2020; Vogel and Hickel, 2023).

Wherein economic growth remains the primary goal, mainstream economists often argue that shared resources will be overexploited due to greed – this is referred to as the 'tragedy of the commons.' They propose privatisation as the solution - selling the shared resources to private owners, who will manage the commons to maximise its output and their profits. An emphasis on privatisation has led to the financialization of goods and services necessary for everyday living such as access to food, healthcare and banking services. It has also led to the 'responsibilization' of the individual e.g. accessing basic goods is increasingly an individual's own responsibility and decreasingly the responsibility of the State (Finlayson, 2009; Bryan and Rafferty, 2014; Crouch, 2009; Beck, 2007).

Basic services such as utilities, food and care services were historically 'low risk, steady return with long time horizons and expectations of a 5% return on capital' (Earle et al, 2018: 43). However, increases in privatization, outsourcing, and private sector financialization (e.g. fund investors) have led to the requirement for returns of more than 10% on investments in the provision of basic services. To facilitate higher returns to shareholders - various instruments are used – e.g. investment rationing, tax avoidance, asset stripping and loading enterprises with debt; jobs are structured such that wages are low and precarious contracts offered; and suppliers (e.g. farmers) are underpaid. These activities have resulted in privatised gains and socialised costs (Russell et al, 2022: 1072).

To address the environmental and social justice harms produced by the prioritisation of economic growth Kate Raworth (2017) proposed the concept of **Doughnut economics**.

> The Doughnut consists of two concentric rings: a social foundation, to ensure that no one is left falling short on life's essentials, and an ecological ceiling, to ensure that humanity does not collectively overshoot the planetary boundaries that protect Earth's life-supporting systems. Between these two sets of boundaries lies a doughnut-shaped space that is both ecologically safe and socially just: a space in which humanity can thrive (DEAL, ND)

Advocates of doughnut economics are critical of green growth approaches which maintain a capitalist, for-profit, extractive system. They argue that, currently, due to extreme power differences between sellers and buyers in markets for essential goods such as energy services, sellers can demand high prices and earn high profits. They also argue that unregulated or poorly regulated markets are leading to environmental harm and destruction. They criticise how businesses which produce high emissions and damaging waste often do not pay for the costs of their practices to planetary ecosystems (Regenerative Economics, ND).

Decoupling GDP from material use and emissions is needed but distinctions can be made between:

- Relative decoupling – a decline in the material/carbon intensity of GDP i.e. GDP grows faster than material use and emissions
- Absolute decoupling – GDP grows while material use and emissions decline
- Sufficient absolute decoupling – GDP grows while material use and emission decline to the degree necessary to keep the economy within planetary boundaries

As Kallis et al (2025: 65) note, the 'consensus from recent reviews and meta-analyses is that while relative decoupling of GDP from material use is common, there is no evidence of sustained absolute decoupling. Moreover, modelled projections indicate that at the global scale, absolute decoupling is unlikely to occur even with optimistic assumptions about technology.' From the perspective of Doughnut economics, the solution is to replace the goal of increasing GDP with the goal of improving wellbeing within planetary boundaries. Raworth (2017) argues that while degrowth - understood as a restructuring of systems which leads to significantly reduced resource and energy use - is necessary, it needs to be targeted; and selective growth is needed ensure basic standards. For example, many countries in the Global South need to grow aspects of their economy to ensure people's basic needs are met; some sectors need to grow worldwide (e.g. forestry); and other sectors need to shrink (e.g. meat production).

Doughnut economics proposes that our understanding of 'the economy' must be regenerative and distributive by design. All economic activity needs to take place within 9 planetary boundaries and after providing for 12 basic human needs i.e. people have rights to basic social services without overshooting ecological boundaries.

Table 3.1: The Ecological Ceiling and Social Foundation of Doughnut Economics

Planetary Boundaries – Ecological Ceiling	Human Needs – Social Foundation
Climate change	Energy
Ocean acidification	Water
Chemical pollution	Food
Nitrogen and phosphorus loading	Health
Freshwater withdrawals	Education
Land conversion	Income and work
Biodiversity loss	Peace and justice
Air pollution	Political voice
Ozone layer depletion	Social equity
	Gender equality
	Housing
	Networks

Source: (DEAL, ND)

Doughnut economics divides the economy into four provisioning systems – Household, State, Market and the Commons. The goal of measures designed for such provisioning systems is to guarantee basic social services for all whilst facilitating absolute decoupling of GDP from nonrenewable resources or transitioning to a low or degrowth economy. In recognizing the societal contributions of activities in the household, by the State and through the Commons - which are currently calculated as costs or free by GDP metrics - doughnut economics places value on social and ecological needs and limitations (Regenerative Economics, ND).

The **Foundational Economy Collective (FEC)** is a group of Europe wide academics who propose that the core 'policy question is not about the technically correct economic policy but about how to do foundational politics so citizens can access quality foundational services provided by decently paid workers' (Earle et al 2018: 41-42). Similar to doughnut economics the FEC seek to make visible those contributions which are invisibilised and/or accorded minimal value within mainstream economic theories. The 'economy' is thus segmented into four parts

- The **core economy** is defined as 'noneconomic' as it is our relations of care and connectedness;
- The importance of the **foundational economy** is stressed - and divided in two parts
    - **Material** foundational economy – e.g. utilities (water, energy), agriculture and food, transport, housing
    - **Providential** foundational economy – e.g. education, healthcare, elder care, social care;

- The **overlooked economy** is one-off, culturally specific purchases necessary for a good life;
- The **tradeable competitive economy** – other goods and services e.g. high end electronics, new cars (Foundational Economy, 2018)

Four basic principles are proposed for the foundational economy:

1. The integration of economic and social policy which will require rethinking the objectives of policy. For example, while higher productivity is relevant to adult care, it cannot be the main objective of policy if care is about maintaining the social relations of older people as much as about meeting their medical needs.
2. Accept that policy means learning from experiments and social innovation which in the first instance will be local. For example, there is a need to experiment with policies to secure the future of grounded, family owned Small and Medium Enterprise (SME) firms; and adapt training and business assistance models for micro firms
3. Accept innovation will not abolish hard policy choices. Many sectors of the foundational economy are built on the exploitation of cheap labour; new policy objectives and instruments in sectors like care will not increase wages unless consumers or taxpayers pay more
4. Recognise the need for participative deliberative democracy rather than top-down policy with government as the leading actor. Coalitions of regional and local actors, with intermediary institutions often in the lead and government in an enabling role could be experimented with (Earle et al, 2018: 43)

Both advocates for doughnut economics and the foundational economy are critical of the use of GDP as the key measure of economic success. A focus on growth makes high-growth sectors (e.g. tech) attractive to policy makers who want to achieve conventional GDP targets. It does not orientate them to think of local liveability in terms of access to good quality basic services; the creation of decent work and quality jobs; and ensuring businesses deliver social benefits to local communities (Wahlund and Hansen, 2022: 178).

A long-time criticism of GDP is that it excludes everything that cannot be quantified in monetary terms and values items that can be quantified in non-differentiated terms. This leads to contestable values and omissions. For example

> items like domestic labour or environmental costs have to be omitted insofar as they do not have market price tags; quality improvements are difficult to measure; items like arms production are included though they contribute little to welfare; measures of finance sector output are contestable; public sector non- market output is entered at cost (Froud et al, 2018: 5).

GDP growth does not necessarily improve living standards for many with wealth and income inequality increasing over the past 50 years (Earle et al, 2018; Kerr and Vaughan, 2024).

Governments increasingly acknowledge the need to shift to a plurality of indices that account for the social and environmental impacts of economic activity. However, according to the FEC this does not address two more fundamental problems with a reliance on GDP metrics:

1. The privileging of the GDP and GVA number encourages an understanding of the economy as productionist - where activity generates earned income from providing goods and services with use value. This view has its origins in the context of the depression of the 1930s/war time economic management and has questionable relevance in the context of present-day financialised capitalism where 'what might be called the rentier circuits of wealth accumulation and household balance sheets are important and need to be integrated into any account of regional income differences' (FEC, 2018: 5). Raworth (2017) makes a similar point arguing that assets such as equity shares, certain types of land, education, and patents are the **fundamental generators of inequalities**. A focus on income inequalities, separate from asset inequalities, overlooks the root causes of inequalities. Raworth thus focuses on the redistribution of control over these assets as central to addressing inequalities
2. The problem not only relates to what is included and accorded value, what is measured as a cost and therefore not 'productive', and what is excluded. The problem is that the economy is made up of heterogeneous and incommensurable objects that cannot be simply added up together. For example, affordable healthcare makes a very different kind of contribution to the overall 'economy', than an inexpensive fast-fashion dress, the production of which generates significant social and environmental harms (Froud et al, 2018: 5-6)

The FEC proposes a metric based on households' access to foundational provision.

> the primary concern of economic policy in every region and national economy should always be with the adequacy, affordability and continuous supply of foundational daily services because housing, health care and utility supply are prerequisite for the well-being of every citizen in every household in the polity (Froud et al, 2018: 8)

To track levels of foundational liveability the FEC propose that partial measures can be obtained by 'working down subtractively from gross income and observing how tranches of income in different types of households are spent on various objects necessary and discretionary, foundational, overlooked and competitive' (FEC, 2018: 8). In economies which support foundational liveability all households, regardless of income or wealth, would have access to good quality, affordable, accessible and reliable basis goods and services.

In common with advocates for universal basic services, foundational economy advocates argue that democratic control over and access to high quality basic goods and services, and improved working conditions in these sectors, are the most effective way of addressing persistent inequalities. Access to services such as healthy and high-quality food, public transportation, education and healthcare enable people to fulfil their potential.

The distinct understandings of 'the economy' within each of the economic pathways leads to distinct priorities for the economy. Continued growth in tradeable goods and services through financialisation (which is assumed to benefit all) is the priority of a green growth pathway. Social and ecological needs are prioritized within doughnut economics because there is an understanding that the economy needs to be regenerative and distributive by design. The provision of basic goods and services through place-based strategies of alternative economic development is the priority of advocates for a foundational economy.

Tables 3.2 and 3.3 provide an overall comparison of definitions and priorities across the three economic pathways.

Table 3.2: Definitions of the Economy within the Three Economic Pathways

Green Growth	Doughnut Economics	Foundational Economy
Tradeable Competitive Economy - includes goods and services consumed by all. However these are envisioned as best provided on a for-profit basis. The economy is understood as a singular zone of activity separate from other social relations (environment, care relations, democratic political processes and structures). Economic resources - including 'fictitious commodities such as money, land and labour are considered an end in themselves' (Russell et al, 2022: 1075).	**Four provisioning institutions** identified as part of an embedded economy model (Regenerative Economics, ND)	Divided in **four parts** (Foundational Economy, 2018)
	Household - also referred to as the core economy - members of a household - as a provisioning institution - provide care for each other through tasks such as cooking, cleaning, parenting, planning and organising activities. Largely performed by women and girls, unpaid, and largely invisible to business and governments. This leads to 'gender differences in time devoted to household work and paid income that make women more economically vulnerable'(Regenerative Economics, ND)	**Core economy** - unpaid relations of care and social connectedness e.g. family, friends, communities. Defined as non-economic, necessary for social reproduction.
	As a provisioning institution **the State** supports the household through services such as education, healthcare and social care; through laws such as sick leave, parental leave and annual leave; through infrastructure to support economic activities e.g. roads, electricity grids, water treatment; and through laws such as banning toxic materials, labour laws, support for research and development activities, land zoning e.g. National Parks, farmland (Regenerative Economics, ND).	**Foundational economy**: the 'part of the economy that creates and distributes goods and services consumed by all (regardless of income and status) because they support everyday life' (Bentham et al., 2013: 7). **Material foundational economy** includes utilities - energy, water; transport and distribution; banking; and food production, processing and retailing. **Providential foundational economy** include health and care; education; social housing

	Market - a provisioning institution where goods and services are priced, bought and sold. Markets only value what is priced and only provide goods and services to those who can pay. Unregulated or poorly regulated markets engage in extractive behaviour that widens economic inequality, degenerates Earths systems, and undermines trust and the social cohesion we need to cooperate to reach shared goals (Regenerative Economics, ND).	**Overlooked** economy - occasional purchases of ordinary cultural necessities e.g. new clothes, new furniture, holidays.
		Tradeable Competitive Economy - cars, electronics, holiday homes, investment properties
	Commons – doughnut economics proposes that the commons is, our ought to be, a provisioning institution of free and open resources. The Earth is a source of finite and renewable reources and also a sink for wastes. Commons resources examples include natural resources, cultural resources such as language and rituals, digital resources like Wikipedia. Regenerative economics proposes that people are able to successfully manage shared resources for the good of the community, sustainably and equitably (Regenerative Economics, ND).	

Table 3.3: Orientation, Role of Institutions, and Economic Development Policy Priorities

	Green Growth	**Doughnut Economics**	**Foundational Economy**
Primary Orientation	Economic growth in the 'tradeable goods and services' portion of the economy	Prioritises social and ecological needs. Regenerative and distributive by design. All economic activity needs to take place within 9 planetary boundaries and after providing for 12 basic human needs - people have rights to basic social services without overshooting ecological boundaries.	Building 'socially embedded institutional structures' (Russell et al, 2022: 1075) that provide for people's needs or livelihoods. Success is measured according to metrics such as social value or democratic processes and outcomes. Prioritises wellbeing and social good, not growth.
Role of Institutions	Enable economic growth	Guarantee basic social services for all and facilitate absolute decoupling of GDP from nonrenewable resources *or* transition to a low or degrowth economy	Structure foundational sectors so that they primarily service citizen needs i.e. set limits on growth in sectors which fulfil basic needs - utilities, food, care
Economic development policy priorities	Centred on the potential of 'high-tech, knowledge-based industries, and property-led regeneration to increase GDP' (Wahlund and Hansen 2022). Focused on boosting growth through better transport links, attracting mobile investment to large projects, and training programmes to prepare ppl for the labour market.	Prioritise funding social and green innovations. Redistribute the value of land, money, labour, technology, and knowledge	Focus on place-based strategies of alternative economic development. By differentiating domains of the economy, the foundational economy prioritises the provision of services and goods consumed by all on a not-for-profit basis and within the environmental limits necessitated by climate breakdown.

Relationship of the Labour Market to Social Supports

As outlined in D2.1 facilitating labour-market transitions is a key element of the just transition. However, the interconnections between the welfare state and labour market policies are ill-defined and the role of the welfare state in responding to the effects of the twin transition on the labour market is not well defined (Galgóczi, 2023: 66). Additionally environmental concerns have largely been absent as a consideration in social policy making until very recently. This is despite a significant and growing evidence base that the dominant understanding of the complementarity of the welfare state and continued economic growth are unsustainable from an ecological perspective (Koch 2022, Hirvilammi, 2020).

The **green growth** pathway maintains a clear division between the economy of tradeable goods and services and all other spheres such as the environment, social supports and democratic systems and processes. Social supports are calculated as a cost within key metrics such as GDP. Welfare provisions are primarily designed to incentivise participation in the labour market and the financial resources for public social expenditure are linked to, and dependent on, continued economic growth (Galgóczi, 2023: 68; Mandelli, 2022). The 'separateness' of the economy from all other provisioning systems has led to significant challenges including:

- Wage stagnation and income polarisation
- Informal and precarious work contracts
- Job losses due the twin transition
- Unaffordable housing, healthcare and education
- Unaffordable low carbon technologies
- Energy and transport poverty
- Gender and Race gaps
- Growing regional disparities
- Vulnerability to environmental hazards and adverse climate events (floods, heatwaves etc) (Galgóczi, 2023)

In the green growth paradigm social entitlements are funded through income and consumption taxes. As such basic welfare services are reliant on a growing economy, and there is no incentive to envision a fundamental restructuring of the divisions between the state, civil society and the economy. Social needs are not adequately provided for by government and businesses, and social innovation is left to an underfunded and ill-equipped third sector (Unger, 2015: 236-238 cited in Russell et al, 2022: 1076).

Doughnut economics is structured as a high-level framework for conceptualizing the economy in a way that accounts for both basic social needs and ecological limitations. It maintains a consistent focus on embeddedness of the economy within larger social and environmental systems. The emphasis on 'the household' and 'the commons' as two provisioning institutions enables a 'move from siloed to complex system thinking. Especially, it enables us to see the interlinkages and broader implications of specific policy proposals on different spheres of the economy' (Wahlund and Hansen, 2022:182). Raworth (2017) argues that the mutual relationship between reproductive work and productive work needs to be considered when designing economic policy.

The **foundational approach** is more pragmatist and prioritises concrete policy instruments which are designed to enable the sustainable development of foundational sectors in the interests of all in a locality. Central to this is the proposition of a **social licensing system**.

The FEC argue 'that policy makers need to balance concern with jobs and wages with more attention to the essential goods and services like housing, utility supply, health, education and care. These are not individual consumption from income but collective consumption because they depend on social provision of the foundational reliance systems which keep us safe and civilised' (2020: 3).

They propose social licensing as a regulation which places 'social and environmental obligations on all corporate providers of foundational services' (FEC, 2020: 8). Social licensing rests on the principle that since foundational enterprises are given **public privileges** (e.g. regulated catchment areas for supermarkets) they should have corresponding **social duties** (e.g. living wages, fair treatment of suppliers, payment of taxes, limiting the use of debt finance, support for community activities).

> Profit and not-for-profit foundational providers in effect have a territorial franchise through their networks and branches; quid pro quo in return they should offer something social in return, like ending tax abuse or insecure employment. All other large corporates should be brought into this regime as and when they want anything from government (e.g. bailouts, planning permission, government contract, training etc.) We recommend social quid pro quo because narrow economics-based regulation of competition and markets to protect the consumer has failed and is increasingly irrelevant in a platform economy (FEC, 2020: 9).

The FEC propose that all material and providential foundational activities should be located in the public domain regardless of the ownership structure of the entity providing the goods and services. This is not to propose that such enterprises are owned by the State but rather that their status is changed, such that they have the same obligations as public bodies (Wahlund and Hansen, 2022: 174)

Table 3.4 outlines the understanding within each of the three economic pathways of consumption patterns and the environmental costs of economic activity and consumption patterns. Table 3.5 outlines the understandings of costs, wealth and value within each economic pathway.

Table 3.4: Costs of Economic Activity and Consumption Patterns

	Green Growth	Doughnut Economics	Foundational Economy
Environmental Costs of the Economy	Environmental costs of economic activities are largely unaccounted for – green growth establishes 'sacrifice zones' primarily in the Global South and subsidises private corporations without imposing substantive environmental or social obligations. For example fossil fuel energy providers do not pay 'for externalities such as the long-term storage of their waste product CO2, the damage already incurred because of CO2, other wastes such as SO2, and they also receive some of the highest subsidies globally – far more than low-carbon energy resources' (Heffron and McCauley, 2017: 665)	Environmental costs of economic activities are accounted for through the 9 planetary boundaries that cannot be exceeded - climate change, ocean acidification, chemical pollution, nitrogen and phosphorous loading, freshwater withdrawals, land conversion, biodiversity loss, air pollution, ozone layer depletion.	Environmental costs are acknowledged but not properly accounted for - recent FE literature has referenced the environmental costs of the provision of services and goods for all, and the need to ensure such provision is within environmental limits, but this is not yet a core concern. Key criticisms of the foundational economy approach are that it is western-centric, insufficiently account for ecological costs, and does not have priority-setting mechanisms for navigating competing social and environmental aims (Hansen, 2022). Further attention needs to be given to democratic governance modes that fully incorporate ecological limitations (Barnthaler, Novy and Plank, 2021; Calafati et al 2021).
Consumption patterns	Private consumption based on market incomes e.g. high income earners are able to afford quality services	In line with arguments for increased collective provision of services and restructuring consumption of services by switching from high- to low-carbon goods and services (Gough, 2022: 465).	Emphasis on the collective provision of accessible and affordable basis services – material and providential foundational economy. Focus of analysis has only recently turned to the need to transition from high-to low-carbon goods and services
Environmental costs of consumption patterns	Largely unaccounted for e.g. subsidisation of fossil fuel energy providers. Private consumption accounts for around 60% of GDP and a higher share of	Environmental costs of consumption patterns are accounted for through the 9 planetary boundaries	Sectors within the **providential** foundational economy such as education, health and care are low carbon sectors as they tend to be less resource intensive and less dependent on long-distance supply chains. However sectors within the **material** foundational

	Greenhouse Gas Emissions (Gough, 2022: 465).		economy such as agriculture and food, energy, mobility and housing are some of the largest emitters of GHGs. Much work remains to be done on how a social licensing system could 'reconcile the meeting of foundational needs with environmental sustainability, particularly when it comes to regulating and controlling foundational sectors dependent on long-distance supply chains' (Wahlund and Hansen, 2022: 182).

Table 3.5: Understandings of Costs, Wealth and Value

	Green Growth	Doughnut Economics	Foundational Economy
Sources of Income/Wealth for Individuals	Wage Labour/Generational Wealth	Wage labour. Also proposes reform of ownership of land, money, labor, technology, and knowledge to balance incomes	Wage labour – prioritises achievement of improved working conditions and good wages in the material and providential foundational economy.
Sources of Revenue for Governments	Income is taxed; wealth taxes are minimal	Proposes the introduction of global corporate taxes to increase revenues for public services in developing countries and to distribute incomes from wealth.	Proposes taxes on wealth to fund and expand on services in material and providential foundational economy. Proposes progressive income taxes and a graduated land tax that 'shifts the local tax burden from lower valued to higher valued properties, which makes sense from a foundational perspective as higher valued properties are generally centrally located with good access to foundational services such as transportation, healthcare, schools' (Wahlund and Hansen, 2022: 180)

Care Economy	Not seen as productive. Little monetary or symbolic value assigned to care.	Accounted for through an understanding of the household as a provisioning institution. Emphasises that the mutual relationship between reproductive work and productive work needs to be considered when designing economic policy (Raworth 2017)	Foundational economy literature prioritises the provision of high quality social, health and education services thus assigning them monetary and symbolic value. However currently not much attention is paid to what they define as the 'core economy' which is defined by relations of care and connectedness (family, friends, volunteer community work) and as such is unpaid/unwaged care work (mostly provided by women) i.e. the impact of gendered norms of care work provision is inadequately accounted for.
Basic services provision	Universal provision of education is available in EU countries to second level. Provision of some level of universal healthcare services also available. Complementarity of the welfare state and economic growth is aimed for – additional welfare provisions are primarily designed to incentivise participation in the labour market, and the financial resources for public social expenditure are linked to economic growth (Galgóczi, 2023: 68; Mandelli, 2022)	Argues for provision for 12 basic human needs as the social foundation – water, food, health, education, income and work, peace and justice, political voice, social equity, gender equality, housing, networks, energy.	Links between the FE and arguments for Universal Basic Services (Coote, 2021)

Both doughnut economics and the foundational economy paradigms stress that the current structure of the economy, labour market and social supports generate inequalities and environmental harms. Both approaches are orientated toward developing a redistributive economic model at policy design stage. The argument is that a well-designed redistributive economic model would improve living standards for all. We can see how the mitigation strategies to address injustice and environmental damages are shaped by the underlying understandings of the economy and the function of the labour market and of social supports.

3.3. Means of Addressing the Eco-Social-Growth Trilemma

Reconciling environmental, social, and economic policy goals is challenging, presenting us with what is termed the '**eco-social-growth trilemma**' (Mandelli, 2022). The social and sustainability policy literature emphasizes the importance of systems thinking and transformative action. However, the ability of policy makers to design effective mitigation strategies that enable fair burden sharing and insulate structurally disadvantaged cohorts against the negative impacts of transition policies may be hampered by the institutional structures in their countries (Slevin and Barry, 2024).

Initial research indicates that countries with corporatist institutions and concertation processes are more likely to be able to implement the long-term strategies that transition requires. In contrast, more liberal market economies that foster interest pluralism are less likely to be able to set a 'steady long-term transition policy path' (Petmesidou and Guillén, 2022: 326). The institutional structures could also influence the degree of public support for various eco-social supports (e.g. universal basic services; participation income) and the specific forms best suited to those contexts (Koch, 2022; Laruffa et al, 2022; Krause et al, 2022). Gaining public support for the restructuring of social supports arguably requires medium-term, participatory, deliberative citizen forums (Gough, 2016; Bonvin and Laruffa, 2022).

3.3.1 Dimensions of inequalities at the climate-environment nexus

As outlined in D2.1 there are five main dimensions to inequalities within the climate-environment nexus.

1. Inequalities related to causation/carbon footprint
2. Inequalities related to the effects of climate change – e.g. impact of extreme weather events by geography, by gender, by migrant status, by age.
3. Inequalities related to adaptation capacities – e.g. protection against extreme weather events, access to housing and food supply
4. Unequal exposure to environmental hazards and toxic substances
5. Unequal impact of necessary climate policy measures e.g. distributional effects, employment effects (Galgóczi and Akgüç, 2021)

Depending on the focus of politics and the structure of funding mechanisms, climate policies will have differential distributional effects and differential employment effects. Examples of differential distribution effects include the regressive effects of carbon taxes on low-income households; low carbon products' high upfront costs disadvantaging low-income households; and, depending on its structure a wealth tax would negatively impact the wealthiest top 1-5% of the global population.

Examples of differential employment effects include how a focus on trade liberalization rather than trade regulation in free trade agreements risks rights violations (e.g. labour rights violations and forced community displacement in mining regions); a continued absence of focus on the care economy means that gender inequalities will remain and likely deepen; policies which do not consider local contexts risk negative social impacts e.g. family separation in search of work.

3.3.2 Barriers and enablers of eco-social policy development processes

The challenges facing policy makers are significant. Restructuring the economy and social supports would need significant public support, the beginnings of which could be achieved with citizen assemblies for example.

The combined arguments of doughnut economics and the FEC suggest that policy makers need to formulate programs for foundational provision on a **sustainable, multi-scalar basis**, considering the **social and ecological dimensions of the local-to-global interlinkages** of foundational goods and services (Wahlund and Hansen, 2022: 183). At local and regional level multiple projects across the EU are already engaged with local stakeholders, they have identified key requirements of a just transition underpinned by human rights and they have developed achievable goals, often adapting elements of alternative economic paradigms to suit local conditions.

The transition will require **supports for affected workers** to be put in place. Yet the literature demonstrates that the current structure of welfare provisions, wherein social policy measures are dependent on continued economic growth, is inefficient and in fact contributes to climate breakdown (Gough, 2017; Hirvilamni and Koch, 2020; Hirvilamni, 2020). As such the configurations of social supports are also in need of transformation as part of the twin transition to 2050. This is not properly recognized within the green growth paradigm. It is partially recognized through the foundational economy approach and fully acknowledged within doughnut economics.

As outlined in D2.1 (pg 63-65) Gough (2016) argues that a 'business as usual' (BAU) approach in OECD countries would involve the 'layering' of environmental policies on existing social policies leaving the underlying reproduction, production, and consumption systems in place. This further disadvantages existing structurally vulnerable cohorts including low-income households and people without the skills to transition to green jobs, leaving **fair burden sharing** and the **principles of climate and energy justice** largely unaddressed.

An alternative scenario involves stringent emissions reduction programmes integrated with radical social policies including eco-social consumption and working policies. Gough (2022) distinguishes between two scenarios to address climate breakdown within social policy.

1. A substantial increase in green capital spending, with the public provision of universal essential services extended.
2. An *economy of sufficiency.* This recognizes the obligations of developed nations to contribute to **decarbonization on a global scale**. It requires 'tackling consumption patterns that are unsustainable, but to do so in a fair way that preserves consumption of necessities and other activities that enhance flourishing. Such an economy of sufficiency would begin to address the 'ceilings' of luxury consumption, excessive wealth and unproductive labour' (Gough, 2022: 470).

The scenarios developed by Gough (2022: 470) compared to present interventions are listed below:

Table 3.6: Welfare State Interventions: Scenarios 1 & 2

Household sector	Present Welfare State Interventions	Scenario 1 Proposals	Scenario 2 Proposals
Employment	Education training	Jobs-oriented Green New Deal and Universal Basic Services Stimulus	Shrink financial, rentier, luxury, wasteful and unproductive employment
	Activation Programmes	Jobs Guarantee	Reduce hours of work
Market incomes	Minimum Wages	Fair Wages	Fair pay ratios in corporation and other institutions
		Strengthen collective bargaining	Implement ceilings on income
			Redistribute wealth
Disposable incomes	Pensions	GMI - Guaranteed Minimum Income/Universal Basic Income	Guaranteed minimum income
	Other cash benefits		Progressive tax options
	Housing benefit		
Public in-kind benefits	Health	Universal Basic Services Expanded Social Consumption	Further expand public services
	Education	Social Care	Shrink luxury and high carbon consumption
		Childcare	
		Housing	
		Transport	
		Internet Services	

Gough (2022)

Integrated public provision of certain services (scenario 1) is more environmentally sustainable than market provision of the same services due to the greater macro-efficiency and lower expenditure shares, better allocation of resources and better procurement practices (Gough, 2022: 465). Tax financed social consumption of services is inherently redistributive and reduces inequality, automatically targeting 'lower income households without the disincentive effects that often result from money transfers' (Gough, 2022: 465; Verbist et al 2012). The positive redistributive effects, and reduction in energy and emissions, case is strong for collective provision of services such as health, education, housing, transport, internet and care – and this is now recognised by climate science (Verbist et al 2012; Ivanova et al., 2020; Millward-Hopkins et al., 2020)

Universal basic services would raise the % share of the **public consumption** of services. There can be no standard way to implement Universal Basic Services, implementation can involve public and private providers, but '*entitlements* to certain levels of provision can be guaranteed and these can be backed up by a menu of public interventions, including regulation, standard setting and monitoring, taxation, and subsidies' (Gough, 2022: 465).

However universal basic services are insufficient to address emissions. Private consumption accounts for around 60% of GDP and a higher share of Greenhouse Gas Emissions (Gough, 2022: 465). As such, achieving '1.5-degree lifestyles' in developed nations calls for significant restructuring of household consumption patterns. Both lower limits of consumption that guarantee basic needs provision for all, and upper limits to consumption in domains where such limits would enhance essential freedoms are necessary (Sahakian et al, 2021)

Achieving an economy of sufficiency (scenario 2) would involve a rethinking of consumption patterns. This necessitates moving away from seeking to maximise our individual consumption of goods and services, to enabling sufficiency for all, in the context of climate breakdown.

> Minimum consumption standards will ensure that individuals living now or in the future are able to satisfy their needs, safeguarding access to the necessary quality and quantity of ecological and social resources. Maximum consumption standards, in turn, are needed to ensure that consumption by some individuals does not threaten the opportunity for a good life for others. The space between the floor of minimum consumption standards and the ceiling of maximum consumption standards produces a sustainable consumption corridor. It is the space within which individuals may make their consumption choices freely and sustainably. It is where they have the freedom to design their lives according to their individual notions of a good life (Fuchs et al, 2021 cited in Sahakian et al, 2021: 308)

Policy makers arguably need to consider how to place limits or ceilings on individual consumption patterns.

Income and Wealth	**Consumption**	**Labour**
Riches	Luxuries	Unproductive
	Ceiling (above which Surplus)	
Prosperity	Comforts	Conventional
	Floor requirements	
Decent Minimum	Necessities	Essential

Figure 3.1: Floors and Ceilings in Three Domains
(Gough, 2022: 467)

The understanding that sustainability and social justice objectives require limits to growth and significant changes to mindsets, lifestyles and the economy is not new. As Sahakian et al (2021: 308) argue, similar concepts include the 'environmental space' (Spangenberg, 1995), 'safe operating space' (Rockstrom et al, 2009), Raworth's (2017) 'doughnut economy' and the 'corridor of sustainable prosperity' (Sachs, 2007). The idea of 'consumption corridor' shares much with alternative concepts but it explicitly takes consumption as the object of study and provides a justification for 'minimum *and* maximum consumption standards via the good life and opportunities for all individuals to satisfy their need' (Sahakian et al, 2021: 308).

Even within a green growth paradigm policy makers need to consider restructuring consumption patterns if significant reductions in GHG emissions are to be achieved in line with the principle of fair burden sharing

3.3.3 Participatory policy design, implementation and review

Operating within a sustainable 'consumption corridor' includes 'non-essential' but socially important jobs and consumption e.g. home improvements, holidays, festivals, nightlife social activities etc. However, a shift to eco-social consumption and working patterns would be a radical change and would challenge dominant interests and ideas, for example 'consumer sovereignty' and unquestioned economic growth. It would need to be accomplished in the face of accumulated policy legacies of existing welfare and environmental states. A 'necessary precondition would be *extensive consensual policymaking* involving key constituencies of interest to set the frameworks for markets' (Gough 2016: 42, our emphasis).

The foundational economy literature is focused on developing pragmatic enablers of social policy development that gives due regard to environmental constraints. FE advocates understand politics as local and stress that 'foundational provision is grounded in territories, hence the importance of plans which **engage local specifics** and are developed with input from citizens using innovative ways of engaging them' (FEC, 2020: 10). To note, the focus on local place-based policy development risks identifying the local as essentially virtuous and abdicating critical multi-scalar (local to global) thinking. However, as Russell et al (2022: 1078) note, the focus on the local is not necessarily such an abdication of multi-scalar thinking, but rather a strategic wager that the local scale offers the best potential for **constructing transformative subjectivities and systems.** Social licensing would function as a mechanism of redistribution, redirecting profits from shareholders and to communities in the form of fair wages and working conditions and accessible and affordable foundational goods and services.

Constructing transformative subjectivities and systems could involve:

- Accounting for and repairing environmental harms (Heffron and McCauley, 2017)
- Accounting for unpaid reproductive activities (Hirway, 2015; Coscieme et al 2019)
- Reconstructing the care economy to address injustices. Defining care work as green jobs (Dukelow et al 2024; Culot and Wise, 2022; Klein, 2022; Littig, 2018)
- Recognising that a range of activities, such as biodiversity restoration or community work should be recognized as beneficial for climate action through, for example, a 'green conditional basic income' (Bohnenberger, 2020: 596) or a participation income, originally proposed by Atkinson (1996), with recent conceptual developments linking it with the eco-social policy paradigm.

The foundational approach highlights the limits of a reliance on innovation to solve policy problems as this 'has often been translated into policy emphasis on certain traded activities, in particular knowledge-intensive business services and high-tech industries' and led to a corresponding *reduction of supports for foundational industries* (Wahlund and Hansen, 2022: 178, our emphasis). That said, Russell et al (2022: 1081) argue that 'foundational thinking needs to engage with the knowledge economy in its broadest sense, most obviously (but not limited to) questions of the production and ownership of data and its role in the governance and reproduction of space.'

Raworth (2017) propose more high-level instruments to organize ownership and control over wealth (enablers of eco-social policy development). These include:

- Cooperatives and other democratic business models that provide people with greater control over wages and working conditions
- Redesign of investment such that it offers returns to shareholders but not at the expense of human rights, community wellbeing and environmental standards. By redirecting credits and savings into productive investments that deliver long-term social and environmental values, Raworth argues that more money could be designated to long-term collective investments such as affordable, carbon-neutral housing and public transport infrastructure (Wahlund and Hansen, 2022: 179)
- Enforcing of environmental and market regulations
- Supporting innovation in sustainable and equitable goods and services
- Empowerment of citizens to engage in democratic progress
- Access to knowledge resources through open-source platforms

To implement a 'just' transition, policy makers must have a clear understanding of the definition of the economy they are using (and thus define what is of value), the relationship between the labour market and social supports, and the scope of the proposed mitigation strategies to achieve social justice, green and digital transition objectives. They could then situate the eco-social policy design on a reactive-preventative- transformative continuum (Mandelli, 2022).

4 FITTER Approach

4.1. Inequalities, structures and policies

As outlined in D2.1 FITTER does not anticipate that a just transition to a carbon neutral economy is possible without a fundamental restructuring of economic models, land stewardship, energy sources, and mindsets. We propose situating the 'eco-social-growth' trilemma as a 'systems trilemma' with complex relationships between 'reproduction, production and consumption systems' (economy), 'planetary ecosystems' (environment) and 'political structures and norms' (social) framed in terms of the reality defined by the climate emergency. We propose situating 'justice' at the core of the wider triangle.

Law and policy, guided by justice concerns, should effectively intervene to (re)structure systems so that we remain within planetary boundaries and enable liveable lives for all. Current climate policy is arguably too focused on piecemeal adjustments to the tradeable competitive economy and neglects accounting for transformations needed with regard to our relationship to planetary ecosystems (environment), and political structures and norms (social) - including the possibilities for enabling eco-social political structures and norms.

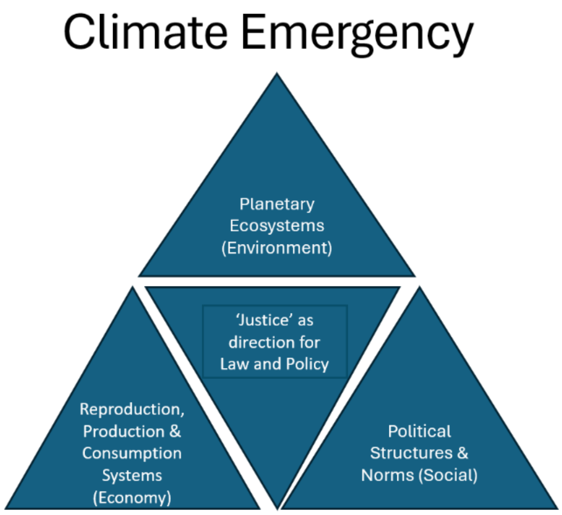

Figure 4.1: The Systems Trilemma
(See also Galgóczi, 2022 and Heffron and McCauley, 2017)

As outlined in D2.1, different theories of justice have different subjects of justice (the unencumbered individual, women and gender minorities, the racialised subject); domains of justice (public institutions, the domestic, the divide between Global South and Global North); and objects of justice (maldistribution of goods or liberties, unequal responsibility for unpaid care-work, historical and contemporary relations of power that oppress (post)colonised peoples).

In D2.1 we defined the just twin transition as a process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account. We proposed taking the 'historically constituted and corporeally vulnerable rights-bearing individual' as the subject of justice; the four

FITTER sectors as they are embedded in global, national, local and household levels of power as the domain of justice; and both *relations of production* (economic-environmental-social) and *mechanisms of distribution* (economic-environmental-social) connected to each of the sectors as the object of justice.

Throughout our work since then, these definitions have been questioned both at team level and in our engagement with stakeholders. For example, our definition of the subject of justice acknowledges how we are all constituted through historical processes (e.g. colonialism, capitalism) and cultural norms (e.g. the primacy of the gender binary) that preceded us. We are corporeally vulnerable in the sense that we are all in relation with, and dependent on, others in one way or another. We engage in giving and receiving care, and we are also dependent on fundamental goods such as clean air, clean water, and food. However, the root level assumptions of subjectivity are only partially questioned within this definition of the subject of justice. It is important that we are conscious of how European definitions of sustainability for example are reflective of our Western understandings of the self/nature relationship, that these may be our starting points, but that there are alternatives to these understandings also.

Reflecting on our initial definitions we have concluded that rather than attempting a harmonization of the different perspectives, we could, for now, 'stay in the trouble' (Haraway, 2016) of the contradictory approaches that feature in the project and subject these different approaches to ongoing reflection.

4.1.1 FITTER project learnings

FITTER has committed to addressing complex issues in four sectors, characterised by multiple dimensions, and impacting on cross-cutting population cohorts as outlined below.

Advantaged and Disadvantaged Cohorts

FITTER has identified key structurally advantaged and structurally disadvantaged cohorts across the four sectors (see D2.1, D3.1).

A key learning thus far is that these cohorts are advantaged and disadvantaged within and through current systems and structures. To note also, structurally advantaged cohorts are more accurately defined as 'temporarily' advantaged because the project 2.7 degree rise in temperature globally will be catastrophic to many and harmful to all. Addressing the root causes of inequality would require climate adaptation and mitigation measures be transformative of the interconnections between the economy, social supports and the environment.

Scope of FITTER

As outlined in D2.1 there are four strands of work across FITTER tasks– assessment of the production of inequalities; forecasting of impacts under three scenarios; systemic design practice (identifying risks and mitigation planning); and the production of tools to assist stakeholders in addressing intersectional inequalities in the context of a just twin transition.

The FITTER project proposes using 3 co-created scenarios to 2050 to estimate industry skills requirements, labour market changes, and the impact on the financial resources of impacted workers. However considerable research (currently underway in each of the sectors to anticipate labour market changes and skills requirements in particular) demonstrates the difficulty in predicting the impacts of the twin transition on impacted workers (Lankhuizen et al, 2022; Vandeplas et al, 2022; Diodato et al, 2023; Paiho et al 2023; Dorantes and Allen, 2024; Villani et al, 2025) Research demonstrates the need for place-based, systems-wide policy analysis and design, based on dialogue and social consensus-building with and by affected communities, in order to enable a just transition (Dekker, 2020; Banerjee and Schuitema, 2023; NESC, 2023). Work completed in FITTER on governance structures and the baseline scenario in each sector has also demonstrated these complexities. It has become clear that the FITTER scenario development process will not be able to provide sufficient detailed knowledge to hypothesise the impacts on workers and future skills requirements in the four sectors within each of the case study countries with a sufficient degree of reliability.

Even with robust hypotheses though, FITTER mitigation measures (for impacted automotive workers, for example) would need to conduct an in-depth analysis of what is already in place with regard to skills development opportunities; labour market structures; income levels and social supports. Each sector is structured in highly complex ways and located within specific governance constraints. Individual policy measures can work to address, leave unaddressed, or further reinforce existing inequalities and high emissions levels. FITTER arguably cannot comprehensively address the complex ways inequalities are reproduced in the labour market or even in a well-defined subsection of the labour market. There is insufficient time and resources allocated to the case study teams to do the required in depth studies.

While the labour market impacts are very difficult to predict, the baseline scenarios demonstrate that the *consumers* disadvantaged by twin transition measures are low-income population cohorts and/or people at risk of poverty in every sector. High income population cohorts are consistently advantaged within current systems and structures (e.g. they can afford to buy an electric car, retrofit their house and buy good quality food). This evidence is not a novel contribution as it is broadly addressed in research and EU sectoral reports. Research also shows that the digital divide (digital skills or lack of them) is one of the strongest indicators of vulnerability, both for consumers and workers (see D3.1).

An exception to the above findings is the agriculture sector. Work completed for D3.1 shows that energy poverty, transport poverty and housing poverty emphasise structural vulnerability on the final users' side but the current structure of the agriculture sector particularly disadvantages small farmers, young farmers and the female workforce.

As outlined in a recent report on a just transition for the agriculture sector in Ireland (NESC, 2023) the agriculture sector is unique in several ways:

- The agricultural and land-use system is made up of thousands of often small or medium size enterprises. This underlines the importance of understanding and incentivising behaviour, and of clear and consistent communications in supporting transition.

- The transition will principally rely upon thousands of individual farmers innovating and changing practices or land use, across diverse soils, land types and production systems. The active participation and engagement of farmers is therefore of fundamental importance to effective transition.
- The transition is about knowledge, capacity and behaviour change as well as technology and production practices.
- Finally, unlike other sectors, the agriculture and land use sector is unique in the area of climate action as it is both a **source** of GHG emissions as well as a **sink** – capable of removing GHGs. This provides a significant opportunity (NESC, 2023: 37)

To address these findings/learnings four areas of focus have been chosen - renewable energy sources, electric mobility, housing climate isolation, and the digitalisation of agriculture. These elements of the *digital* transition are each identified as key enablers of the *green* transition by the EU. The degree to which they can also enable a *just* transition is thus in need for further analysis. The areas of focus have been chosen as a result of an in-depth analysis of the EU Fit for 55 package and related reports (conducted as part of T3.1).

To structure the work, a core research question will guide the work of FITTER country case study teams. Each question will:

1. Identify key element of the digital transition
2. Focus on the degree to which it supports a wider social objective (i.e. addresses the reproduction of inequalities for either consumers or workers) - and critically assess whether or not it achieves stated environmental targets (green transition)
3. Align analysis with justice principles and dimensions

To best utilise case study team resources, the transition to **renewable energy sources** will be examined in each case study country. Case study teams will also examine one other sector. Teams were allocated a second sector of analysis based on the importance of those sectors to the national economies and partners' existing expertise in these areas.

- Transport – **electric mobility** – Spain; Hungary
- Housing and Built Environment – **housing climate isolation** – Germany; Portugal
- Food & Agriculture – **digitalisation of agriculture** – Ireland; Italy

These amendments are in line with a key element of systemic-design which is that we learn by doing. A case study approach 'can combine empirical research, policy analysis and normative reasoning in order to understand why a policy area generates injustices, and to connect strategies of redress with reasoning and normative principles such as the principle of parity of participation' (Brink et al, 2020: 210). Systemic-design requires openness to reframing problems, accounting for wider systems, and seeking to effectively intervene so that the root causes of the generation of intersectional inequalities is addressed. For an example of the starting point for a case study see Appendix B

The Social and Political Context of Technological Development

A key learning from analysis of twin transition processes has been that technological developments (digital transition) do not necessarily support the achievement of environmental objectives (green transition) and/or social objectives (just transition) (Gao, 2024; Johnson et al, 2020; Bendetti et al, 2023). There is a risk that twin transition policies in Europe could be an example of policy 'running ahead of theory and evidence' (Faggian et al., 2024; Kovacic, 2024).

To address this finding the scenario building process is designed to produce robust hypotheses on technological developments relevant to each area of focus; and these developments will also be further situated within possible political and social contexts to 2050 (T3.2). Subsequent work on risks and mitigation measures will include an analysis of the social and political contexts of technological developments.

Limited Quantitative Measures

A major challenge for reaching FITTER research objectives is the limited availability and quality of quantitative measures. There are three elements of the issue of incomplete data/knowledge.

- The first is *ontological* – the nature of wicked policy problems (i.e. eco-social policy issues) is such that there are no criteria which enable one to prove that all (or even a good range of) solutions have been identified and considered (Crowley and Head, 2017)
- The second is *epistemological* – our abilities to understand eco-social policy problems are limited. This does not mean that we should cease to try to understand them, but that we should do so with humility, acknowledging the limitations of any understanding (Crowley and Head, 2017)
- The third is *practical* – our use of quantitative indicators is dependent on and limited to existing data collection and reporting processes.

The FITTER goal to automate analysis processes needs to be cognisant of the risks of reinforcing exclusions which automation can entail. A high-level review of available indicators conducted within T2.1 has found that there is no set of indicators which can comprehensively address the dimensions of eco-social policy challenges. Furthermore, the literature on risk analysis emphasises that we are living alongside/within conditions of 'deep uncertainty' and risk analyses need to account for this (i.e. they need to be dynamic, adaptive, and equip users with adaptative and systems level thinking) rather than trying to remove deep uncertainty (Cox, 2012; Aven and Zio 2014). Policy makers, disadvantaged groups, and stakeholders cannot provide sufficient data to develop a predictive and quantifiable model of risks and mitigation measures.

To address this finding, FITTER can source indicators relevant to each research question and the findings of the scenario building process. The degree to which they can be used to measure risks, and/or progress (or not) in relation to specific eco-social policy challenges will be assessed and clearly outlined. The indicators can support the development of an automated simplified simulation tool aimed at triggering critical reflection and thinking in policy design.

The Complexities of the Reproduction of Inequalities

The literature review has demonstrated how decision-making powers differ between cohorts. Without ongoing participative policy making processes, large actors can overlook the experiences and needs of structurally vulnerable groups. The reproduction of inequalities and high emissions are also the outcome of place-based systems and structures which reproduce intersectional inequalities (Crenshaw, 1991, Reckien et al., 2018; Rahman, 2018; Collins, 2017; Sultana, 2021). For example, during COVID-19, gender inequalities were exposed albeit not to the same extent: certain women, such as migrant and working-class women, carried a greater burden of reproductive labour (Gilbert et al., 2020; Staniscauski et al., 2020). The pairing 'Disadvantaged Group-Hazard' does not necessarily enable an understanding of the complex processes through which the twin transition is structured and inequalities are reproduced. To partly address this, we have defined hazards as policy measures which produce/reproduce structural inequalities, although these complex interactions are difficult to quantify and incorporate within a predictive system/algorithm and this will be subject to ongoing refinement throughout the project.

Structural Inequalities and Energy Systems
Studies demonstrate complex relationships between inequality grounds and energy poverty. For example, in their analysis of ethnicity and domestic energy injustice, Bouzarovsky et al. (2022) examine long-established links between low quality housing, overcrowding and fuel poverty among households with ethnic minority background. This is in line with wider social concerns to include patterns of social exclusion along ethnic and racial lines. Traditional approaches to energy poverty tend to operate within the classic triad of low income, poor energy efficiency of the home, and high prices of energy. However, this approach was criticized by Grossman and Kahlheber (2017) in their study on energy poverty in Germany, in which an intersectional perspective was adopted. As they argued, the triad only partially explains energy poverty experienced by households as it 'fails to engage with deeper systemic structures and inequalities that underpin domestic energy deprivation.' In the case of Germany, they found that age and health constitute important factors. Illness can trigger energy poverty as more heating is required; at the same time, illness can be a consequence of energy poverty (resulting in a 'vicious cycle' of energy poverty). Those who need to spend more time at home – older citizens as well as parents of young children – tend to have higher energy bills. For single parents with small children this often coincides with low-incomes. Housing market discrimination is also a contributing factor, with low-income families, single parents, or households with migration backgrounds often facing discrimination on the rental market and often forced to housing with low energy efficiency. Migration background was identified as an additional barrier when language barriers were an issue with individuals not being able to effectively communicate with energy providers after falling into difficulties with payments. Severe energy deprivation was found in cases where inequalities intersected: low income, single parents of migration background who needed to spend longer hours at home among the examples provided by the authors of the study. Finally, the study emphasizes that the role of policies in reproducing inequalities need to be accounted for, particularly in relation to housing policies which tend to discriminate against low-income households through housing segregation and low-energy efficient accommodation for families in receipt of state supports.

4.1.2 FITTER methodological development

Based on these learnings we have further developed the strands of work in FITTER (initially identified in Table 7.1, D2.1).

Table 4.1: FITTER Strands of Work: Assessment, Forecasting, Design & Production

Strands and sub-strands		Key Elements	Status
Assessment of the generation of inequalities			
1	Problematise 'vulnerability'	Literature review	Complete (D2.1/D2.2)
2	Develop comprehensive understanding of the reproduction of inequalities in each sector	Literature review Focus Groups	Complete (D2.1/D2.2/D3.1)
3	Assess the degrees to which recognition justice, distributional justice and representative justice are present/absent within the baseline scenario. Develop understandings of inequalities as co-constitutive (not multiple)	Literature review FITTER Labs	In progress (WP3/WP4)
4	Assess where the policy issues fit within wider systems, taking the whole production-distribution-waste mgmt life cycle into account. Identify inequalities in functionings, patterned across groups of people (Intercategorical Approach (McCall, 2005))	Literature review	In progress (D2.2/D2.3/T3.3)
5	Define research question per sector	Literature review	In progress (T2.1), to be reported on in D3.2
6	Identify indicator set which can best address key dimensions of the research question (eco-social policy challenge). Follow Robeyns (2003) five criteria for drawing up a list of functionings to assess - taking all elements of the 'systems trilemma' into account.	Literature review Interviews	In progress (T3.3)
Forecasting Policy Impact (Scenario Building)			
1	Develop scenarios to 2050 situating technological developments within social and political contexts	Interviews Delphi Surveys Focus Groups	In progress (T3.2)
Systemic Design Practice (Identifying Risks and Mitigation Planning)			

1	Consider how to apply justice concerns in practice. Restorative justice includes fair burden sharing, prevention of harm, and repair of harm	Survey of Stakeholders FITTER Labs Computational Analysis	In progress (WP4)
2	Assess where policy design sits on a reactive-preventative- transformative continuum in relation to integrated environmental and social objectives	Survey of Stakeholders FITTER Labs Computational Analysis	In progress (WP4)
Production of Tools			
1	Produce simulation tool to assist policy makers.	Algorithm development/Systems design Platform development API development Validation Pilot Testing Focus Groups	In progress (WP5/WP6)
2	Better Practice Guides	FITTER Labs Literature Review Focus Groups	In progress (WP2-WP6)

FITTER will align with the UN/ILO Guidelines for Just Transition while also highlighting that the evidence base does not demonstrate that a win-win green growth paradigm is sustainable. There is a need for more robust supply side climate policy - prioritising systems change which values the activities of the reproductive sphere and fully accounts for and mitigates the environmental and social costs of extractivism and intensivist production.

The co-creation process seeks to engage specific modalities of power – 'power-with', which stakeholders can enact in solidarity with one another in social movements; 'power-within' which entails the development of a critical understanding of: 1) our own power-within, 2) the forces which shape dynamics of oppression and privilege (power-over) and 3) the possibility of actions (power-to and power-with) (Gaventa, 2021; Rowlands 1997; VeneKlasen and Miller, 2002); and 'power-for' which involves identifying and working for the values and goals that underpin a just transition.

The work is situated in an 'invited space' wherein stakeholders are invited to participate and work through policy challenges posited by project team members. It is primarily situated at the State (national) and Region (regional) levels but could have impact at the EU (supranational) level in the medium term. FITTER does not have influence on hidden power (the ability to control who gets to the decision-making table and what gets on the agenda) nor will it contribute significantly to the shaping of meaning itself (invisible power) as this requires a more long-term piece of work. The work of FITTER is situated within a broader process of social consensus building through which civil society action seeks to change the 'who, how and what' of policymaking (visible power) - so that the policy processes are more democratic and accountable, serving the needs and rights of people within planetary boundaries (Gaventa, 2006, 2021).

FITTER mitigation options and governance recommendations will need to account for:

- Integrated systems-wide policy analysis, design, implementation and review (Blomkamp, 2022). Structural change can be led by governments (upstream interventions), businesses and local authorities providing sustainable options (midstream intervention) and downstream interventions aimed at shaping individuals' decision making (Hampton and Whitmarsh, 2023).
- Addressing justice frameworks and dimensions
- Compiling and critically evaluating indicator sets in line with the work of Robeyns (2003) and Wolff and De-Shalit (2007)
- Understanding of the benefits and limitations of an output focused eco-social policy design (Mandelli, 2022).
- Utilising the six quality criteria for intersectional policy design and evaluation developed by Lombardo and Agustín (2012).

5. Conclusion

The twin transition is a relatively new concept which has mainly emerged in EU policy frameworks post-COVID-19 crisis. While it is assumed that the digital transition will enable the green transition the evidence for this is still relatively limited. Technological advances are reliant on varying levels of resource extraction and energy consumption which must be accounted for in any evaluation of their effectiveness. Some population cohorts are well positioned to benefit from the digital transition in particular but the evidence suggests that other groups may be further disadvantaged by the combined processes of the twin transition. It is notable that existing inequalities are structural (produced by existing systems and structures) and action which is substantively transformative of systems and structures is needed to address the reproduction of inequalities.

In chapter two, we exemplify some of the structural dimensions of existing inequalities in Europe. We also demonstrated that the descriptive analysis of existing inequalities has its limitations. More in-depth understanding needs to be developed in relation to the way in which these inequalities are intersecting across inequality grounds and inequality domains, as well as the root cause of these, with local context also taken to account.

The 'just transition' is usually defined through the principle of 'leaving no one behind'. Yet the potentially affected population cohorts are often vaguely defined. The principle is also problematic considering that most just transition policies are focused on the impact on employment. Much of the focus is on jobs in the 'brown' or 'polluting' industries, which are mainly dominated by male workers. This is arguably a limited and limiting focus for several reasons. First, job losses do not occur in a vacuum and it is the communities, rather than individuals alone, who are affected. In this, it has been argued, women are the often the ones who carry the burden of the structural shift, as they are responsible for household management and budgeting. Second, the focus on 'green' and 'brown' jobs excludes non-polluting sectors such as the care economy from consideration and from accessing supports. There is also a disproportionate focus on the green transition as separate to the digital transition.

In relation to the four FITTER sectors, we argue that the focus on jobs is rather limited, and that both, the employment side, as well as the consumers situation, should be accounted for. In agriculture, we identify gender as the main inequality ground in relation to work, however, inequalities across the food chain should be scrutinized, especially in the context of changing dietary habits. In this case, socio-economic background is most likely to be of importance. In the case of transport, energy and housing/built environment, gender is also highlighted as the main source of inequalities, which is often linked to the male working culture, and also lower proportion of female graduates in the related subjects. In terms of the consumer side, in transport, structural inequalities are often discussed in the context of transport poverty, defined through accessibility, affordability, availability and safety – with gender also featuring as a factor, but also other inequality grounds highlighted, such as disability, ethnicity and race, and the LGBTQI+. Inequalities in energy sector are explored through the concept of energy poverty, defined through the inability to pay bills or inability to keep the house warm. The latter intersects with the inequalities discussed in relation to housing/built environment, usually analysed in the context of housing adequacy, which also contains elements of the energy rating but also is a broader concept incorporating various elements of housing as a human right. The way in which inequalities are explored in these two sectors will require further scrutiny, as similar vulnerable categories – lone mother, elderly citizens, disabled persons and those with migration background – tend to be affected in these domains.

In chapter three we outlined how the need for adaptation to be transformative has been elaborated in conceptual and practical terms. Distinct understandings of the meaning of sustainability, the role of technology in eco-social policy development, and potential avenues of social innovation, result in quite different conceptualisation of the climate crisis and potential solutions or pathways forward. To implement a 'just' transition, policy makers must have a clear understanding of the definition of the economy they are using (and thus define what is of value), the relationship between the labour market and social supports, and the scope of the proposed mitigation strategies to achieve social justice, green and digital transition objectives

In chapter four we outlined key project learnings thus far and further developed the methodology. We have identified significant constraints (e.g. limited quantitative data) and crucial developments (e.g. a team level, in-depth understanding of inequalities and environmental harms as produced through systems and structures).

An eco-social approach to policy design can be built on the understanding that we need to structure adaptation so that it is transformative of systems and enables the wellbeing of all. The evidence base demonstrates the need to transform economies and societies to halt climate breakdown and and promote social inclusion and equality. Analysis of twin transition processes indicate that technological developments (digital transition) can, but do not necessarily, support the achievement of environmental objectives (green transition) and/or social objectives (just transition). It is important that technological developments work in service of integrated eco-social policy design. However eco-social policy design is at a very early developmental stage. Considerable social consensus building processes would be needed to facilitate understanding of challenges we face, and to enable democratic co-determination of the required cultural and societal changes.

Based on work completed thus far the FITTER approach continues to develop. However fundamental elements identified in D2.1 remain core to the research process. These include the importance of

- understanding the reproduction of inequalities and environmental harms as the product of systems and structures
- taking climate, energy and spatial justice into account in policy assessment
- taking a systems approach to policy design, implementation and evaluation;
- enable policy design which explicitly pursues both environmental and social objectives in an integrated way.

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Appendix A

Table A1: Concepts of the Social Contract

Concepts of the Social Contract	Notes
The construct of the 'social contract', understood as a pact between governments and peoples, is central to our understanding of the structure of social and political life. At the centre of the **Western construct of the social contract** is an understanding of the **individual as autonomous, unencumbered, and someone who 'opts in' to the social contract of their own volition**, thereby agreeing to fulfil certain responsibilities, and able to avail of their rights	Scholars have argued that the social contract is based on a **sexual *and* a racial contract**. In liberal theory the domestic realm has generally not been theorised in relation to the pursuit of freedom or to structural constraints on freedom (Kymlicka, 1990). The premises of liberal thoery work to efface the relations of care and the social and institutional structures which produce the liberal individual and enable the social contract. They also render those charged with reproduction, care work and emotional labour as 'subordinate' to the ideal of the rational individual (Squires, 1999: 26-31). Through processes of colonization colonised peoples were subject to genocide, enslavement and segregation and defined as less than fully human. Societies were thus formed through the systematic exclusion of people of colour from contracting and the political sphere (Mills, 1997; Bhambra 2021).
The need to ensure human rights for all (including women and migrants); larger freedom for all including security and protection; and the transformation of economies and societies to halt climate change and environmental destruction are core to the UNRISD **socio-ecological contract**.	An **eco-social contract** arguably provides explicit recognition of three constitutive yet undervalued or ignored elements of the 20th century social contract associated with welfare capitalism – 1) The low to no value attributed to carework (the sexual contract); 2) The legacies of colonialism and coloniality including the present-day structuring of the global economy (the racial contract); and 3) The availability of the planet's resources as raw materials for the production of goods and as a sink for production's waste.

Table A2: The Power Cube (Gaventa, 2006, 2021)

The '**power cube'** developed by Gaventa (2006, 2021) outlines how the interrelationships of spaces, levels and forms of power can be analysed, and the possibilities of transformative action assessed. A policy victory for one group, which fails to take intersecting power dynamics into account, may also reinforce the domination and subordination of others (Crenshaw, 1991).	
Spaces of Power	
Many decision-making spaces are **closed spaces** in the sense that officials	In reality all spaces exist in dynamic relationship to one another; whoever creates the space is

(bureaucrats, elected representatives, experts) make decisions and provide services without the need for broader consultation. **Invited Spaces** are spaces into which people are invited to participate by government, non-governmental organisations, or supranational agencies (Cornwall, 2002). **Claimed/Organic/Third Spaces** are created by less powerful actors out of their common concerns. They may be the outcome of popular mobilisation and involve processes of debate, dialogue and contestation outside of institutionalised policy arenas (Gaventa, 2006; Cornwall, 2002).	more likely to have power within it; social actors may have a lot of power in one space but relatively little power in others; and power gained in one space, through the development of critical consciousness and new skills for example, can be used to enter other spaces.
Levels of Power	
In a globalised world we cannot only focus on power at the local or national level. Gaventa (2006, 2021) outlines four levels of power - **Global** – formal and informal sites of decision-making beyond the nation state; **National** – governments, parliaments, political parties or other forms of authority linked to nation-states; **Local** – subnational governments, councils and associations at the local level; **Household** – the micro-level (2021:14)	Whereas Gaventa (2021: 14) notes that the household level may be outside the public sphere but 'helps to shape' what happens within the public sphere - a feminist lens would argue the inverse – that the public sphere is structured through (and at the same time seeks to efface) relations of domination which are first structured at the household level (Zerilli, 2006).
Forms of Power	
Visible power: observable and recorded decision making. This level includes the visible aspects of political power – the rules, structures, authorities, institutions and procedures of decision making.	Civil society action at this level seeks to change the 'who, how and what' of policymaking so that the policy process is more democratic and accountable, serving the needs and rights of people within planetary boundaries.
Hidden power: agenda-setting. This is the ability to control who gets to the decision-making table and what gets on the agenda. Hidden power operates at multiple levels to exclude and devalue the concerns of less powerful groups.	Civil society action at this level can work to build the power of social movements and new leadership, to influence the shaping of the political agenda and increase visibility and legitimacy of civil society issues, demands and voices.
Invisible power: shaping meaning itself. This relates to the embeddedness of harmful cultural norms which shape the boundaries of what is acceptable.	As Gaventa notes 'processes of socialisation, culture and ideology perpetuate exclusion and inequality by defining what is normal, acceptable and safe. Change strategies in this area target social and political culture as well as individual consciousness to transform the way people perceive themselves and those around them, and how they envisage future possibilities and alternatives' (2006: 19).

Table A3: Modalities of Power

Modalities of Power	Notes
Power over involves situating one group as powerless and another group as powerful. **Power to** relates to our capacities to act.	In reality we are all both powerful and powerless, dependent on wider intersecting power relations. Although empowerment (power to) is a precondition for resistance, it does not necessarily lead to the transformation of relations of domination. We may well be 'empowered' to act in ways that uphold or reinforce relations of domination.
Power-with is understood as the power we can enact in solidarity with one another in social movements. **Power-within b**uilds on the work of Paulo Freire (1997) and refers to the importance of 'conscientisation',	Conscientisation is understood as the development of a critical understanding of: 1) our own power-within, 2) the forces which shape dynamics of oppression and privilege (power-over) and 3) the possibility of actions (power-to and power-with) (Gaventa, 2021; Rowlands 1997; VeneKlasen and Miller, 2002).
More recently Bradley (2020) has conceptualised the modality of '**power-for'**, as referring to 'the combined vision, values, and demands that orient our work.	Power-for inspires strategies and alternatives that hold the seeds of the world we want to create. 'Power-for' provides a logic to transformative power – motivating the sustained movement-building efforts that generate power to, with and within as building blocks for change' (2020: 108 cited in Gaventa 2021: 5).

Table A4: Dimensions of Climate Related Inequalities

Five dimensions to inequalities within the climate-environment-social nexus (Galgóczi and Akgüç, 2021)
Inequalities related to causation/carbon footprint
Inequalities related to the effects of climate change – e.g. impact of extreme weather events by geography
Inequalities related to adaptation capacities – e.g. protection against extreme weather events, access to housing, food supply
Unequal exposure to environmental hazards and toxic substances
Differential effects of climate policies and related inequalities

Table A5: The Triple Injustice of a BAU Approach to Economic Growth

The **triple injustice** of a 'business as usual' (BAU) economic growth-focused approach to policymaking (Mandelli, 2022: 340)	
1. Leaving inequalities unaddressed exacerbates climate change.	Wealthier corporations, nations and individuals remain less accountable (or not at all accountable) for high GHG emissions, unequal societies have low

	levels of cohesion and socio-ecological resilience, and at a global level our collective capacity for action remains underdeveloped.
2. Climate change further exacerbates existing inequalities because	Structurally vulnerable groups are most exposed to ecological risks, and least equipped to deal with climate change induced harms.
3. Climate policies can further contribute to existing inequalities, due to their socially regressive nature	

Table A6: UN/ILO Guidelines for a Just Transition

UN/ILO Guidelines for Just Transition	
A Just Transition should be based on **social dialogue and consensus building**; realise **rights at work**; **promote equitable outcomes,** with a specific focus on being **gender transformative**; adopt a **systems approach to policy design**; create **decent jobs**; take a **place-based approach** to policy design; and **foster international cooperation**.	The agreed ILO Just Transition Guidelines are focused on mitigation and adaptation measures to climate change and climate breakdown. They are thus forward looking and avoid significant analysis of the history of climate breakdown causation. Furthermore, they have been criticised for adopting 'the fallacy' of there being a win-win approach to the Just Transition. Such an approach avoids addressing the scale and urgency of the planetary emergency we face, favouring a green growth paradigm, driven by technology, which keeps us on a collision course with planetary limits (Barry 2012).

Table A7: Environmental, Climate, Energy & Spatial Justice

Concepts of Justice	**Notes**
Bryant (1995: 589) defined **environmental justice** as the 'cultural norms and values, rules, regulations, behaviours, policies and decisions to support sustainable communities, where people can interact with confidence that their environment is safe, nurturing, and protective.'	Although the scope of environmental justice can be global, it is often **focused at a more local level** and concerned with the justice of decision-making on land-use outcomes (Wood-Donnelly, 2023: 142).
As climate breakdown has happened over time - through the cumulative effects of resource extraction and pollution across borders - and the benefits and harms of these activities are inequitably distributed, the **scope of climate justice is global**. A climate justice perspective highlights that climate breakdown is the result of specific norms, beliefs and decisions/actions that are committed to preserving existing social and economic arrangements, and that the	States in the Global North which have contributed most to GHG emissions, and which have benefited from resource extraction from the Global South, are seen as morally obliged to pay more toward environmental protection. The most affected communities by climate breakdown are seen as entitled to special rights and compensation (UN 1992; Stern 2006; Shue, 2010; Skillington 2016).

organisation of social and economic life sustaining those arrangements can be transformed (Skillington, 2012). Principles of right, responsibility, equality and fairness are central to climate justice.	
Energy justice is concerned with developing mechanisms for more just production, transportation, distribution and use of energy.	Heffron and McCauley (2017) take a *systems approach* to energy justice. They place the energy life cycle at the centre of the framework. This includes 1) Extraction; 2) Production; 3) Operation and supply; 4) Consumption; 5) Waste Management. They advocate a three-step analysis of a given issue 1) Analysis of the issue vis-à-vis distributional, procedural and recognition justice tenets; 2) Place the issue in the broader energy life cycle taking global interdependencies and issues into account; 3) Employ applied principles for guidance on practical action and restorative justice as a condition for action.
Spatial justice is concerned with framing the sites and scale of injustices. Its usefulness lies in how it helps identify where injustices are taking place as well as how others beyond the immediate location are impacted. Space can be conceptualised as absolute, relative, and relational at the same time (Przybylinski, 2023: 192).	

Table A8: Dimensions of Justice

Dimensions of Justice	Notes
Justice as recognition - Recognitional injustices include the use of mechanisms of exclusion (ignoring someone, insulting someone, devaluing or dismissing certain perspectives without due consideration); and misrecognising people (drawing on stereotypes rather than lived experiences, failing to unlearn harmful norms).	From an energy justice perspective misrecognition has been shown to occur when local opposition to environmental policies is attributed to ignorance and/or self-interest of specific communities (by policy makers) with insufficient evidence base for such conclusions. Taking the time to understand the core concerns of those who oppose environmental policies, developing a broad based understanding of the assumptions, values and judgements that underpin distinct positions in relation to a policy (in favour, against, undecided) and taking a conflict resolution approach to deliberative public engagement could arguably lead to the creation of a negotiated compromise that does not sacrifice social justice requirements in order to achieve environmental and economic goals (Barry et al, 2008).

Justice as redistribution - An intersectional approach to addressing distributional inequalities proposes that inequalities are structured by norms and societal structures which limit some cohorts' access to both material resources and discursive resources (Fraser, 1995; Crenshaw, 1991). From an intersectional perspective, justice as redistribution does not only relate to a fair distribution of goods and liberties, but also includes a commitment to understanding the histories of discriminatory norms, their material effects, and working to transform both in the direction of more liveable lives for all.	Dynamics of oppression such as sexism and racism (which set limits on the discursive resources of affected communities) underpin and legitimise unequal access to material resources (e.g. gender and race pay gaps).
Justice as representation - When perspectives within a polity are unjustly excluded, silence or ignored there is a risk of 'false negative danger' - where certain public interests are ignored; 'false positive danger' - where common interests are misrepresented and other interests are falsely presented as common interests (Pettit, 2004); and misframing – where the boundaries of the community of concern are drawn to unjustly exclude certain persons (Fraser, 2008). Closely related to the concept of **procedural justice** - the requirement that all stakeholders be included in a non-discriminatory way; that they be able to participate in decision-making; and that government and industry stakeholders commit to participation, impartiality and full information disclosure (McCauley et al 2013).	For example, given that we are embedded within a global economy and addressing climate change as part of a just transition is a collective problem, it would be unjust to ignore human rights abuses in countries wherein raw materials are sourced. Even though it may be unfeasible to recruit participants from affected global communities in direct consultation, policy makers can place a local energy issue in the broader energy life cycle taking global interdependencies and issues into account (Heffron and McCauley, 2017).
Restorative justice involves accounting for and addressing the cumulative impact of past injustices and harms.	

Table A9: Eco-Social Policy Development (Mandelli, 2022)

Eco-social policy development (Mandelli, 2022)	Notes
An output focused eco-social policy has two defining features - first that it **explicitly** pursues both environmental and social objectives, and second that it does so in an **integrated** way.	An integrated policy is based on 'a design in which, first, multiple policy goals can be coherently pursued at the same time, and, second, policy instrument mixes are consistent, in the sense of being mutually supportive in the pursuit of policy goals' (Rayner and Howlett, 2009: 100 cited in

	Mandelli, 2022: 340). Integration thus aims to counter the 'growth-first' logic that remains dominant in the governance of the social, environmental and economic spheres (2022: 345)
Reactive eco-social policies **prioritise environmental objectives** in response to the climate emergency and then **add on a 'social' dimension.** Reactive eco-social policies are developed in immediate response to the climate emergency. These 'reactive' eco-social policies can be further classified as serving either (a) a 'protective' function and so not contributing to growth e.g. transition income support in case of loss of employment in carbon intensive sectors, or income support for energy-poor households; or (b) as serving an 'investment' function and so contributing to growth e.g. reskilling provisions for redundant workers in carbon-intensive sectors (Mandelli, 2022)	Reactive eco-social policies prioritise mitigation and adaptation, e.g. paying the social costs of ecological catastrophes (heatwave, floods etc) or redistributing the costs and benefits of the twin transition.
Preventative eco-social policies **prioritise social objectives** and then **add on an environmental dimension.** They are larger in scale and seek to embed long-term societal transformation. These 'preventative' eco-social policies can also be classified as serving either (a) a **'protective'** function and **not contributing** to growth e.g. long-term income support to allow working time reductions, or (b) a **'preventive'** function and also **contributing** to growth e.g. building energy-efficient social housing, investment in green job creation (Mandelli, 2022)	Preventative eco-social policies can be used to reduce the carbon footprint of a region or nation state, and to promote behavioural change in the direction of sustainable practices.

Table A10: Quality Criteria for Intersectional Policy Design (Lombardo and Agustín, 2012)

Quality criteria for intersectional policy design and evaluation (Lombardo and Agustín, 2012)	
Explicitness, visibility and inclusiveness – it is important to name the operative dynamics of privilege and oppression. If problems are not named, they will not be addressed.	
Articulation – this relates to the ways in which dynamics of privilege and oppression are conceptualised, the degree to which they are explained such that all contributors can engage in productive discussions.	To understand how disadvantages are mutually constitutive (rather than multiple) we need to understand the histories of categorisation. Where no 'explanations or understandings of the nature of the relationship between the categories are expressed' (2012: 489) the dynamics remains inarticulate.

Gendering/Degendering – Lombardo and Agustín argue that keeping gender as a category of analysis (i.e. not omitting gender in consideration of other categories) is a 'quality criterion because it enhances the likelihood that gender equality is treated as an aim in itself, and that this goal is not lost when other inequalities enter the agenda' (2012: 490).	Arguably it is more useful to consider racialised gendering as a quality criterion given the tendency of some uses of intersectionality to seek to adopt a disembodied or 'post-racial' framework (Bilge, 2020: 2317). Adopting a disembodied framework effectively invisibilises processes of racialisation (including the ways whiteness operates as a dynamic of privilege).
Structural understanding of inequalities – to develop transformative policies and address power hierarchies it is important to understand inequalities as systemic i.e. it is norms, structures and institutions which create and reproduce socially stratified population groups.	
Avoid the stigmatisation of specific groups/Challenge privileges – this is linked to developing a structural understanding of inequalities. The naming of groups who experience co-constitutive disadvantages risks stigmatising them as lacking in agency and/or responsible for their own adverse circumstances. To mitigate this risk it is important that practitioners maintain a focus on both oppression and privilege e.g. racism is *productive* of advantages of some people and disadvantages for others. This includes developing a good understanding of the various modalities of power.	Empowerment mechanisms (power-to) which are disconnected from power-with others, power-within or conscientisation, and power-for (as the vision, values, and demands that orient our work), may well lead us to act in ways that uphold or reinforce relations of domination (Gaventa, 2021).
Consultation of civil society in the policy-making process – consultation with a wide range of stakeholders increases the possibility that policy design and implementation adopts a user-oriented approach and incorporates 'a more explicit, articulated, transformative, and less-biased approach to intersectionality since the inclusion of embodied subjects expressing different concerns can promote greater self-reflexivity on one's own biases' (Lombardo and Agustín, 2012: 491).	This does not always work in practice - time, resources, and network contacts can limit the range of stakeholders consulted and the most powerful actors can prevail in consultations (2012: 491).

Table A11: Criteria for Selecting Indicators (Robeyns, 2003)

Criteria for drawing up a list of initial functionings and indicators in a research study (Robeyns, 2003)	
Explicit formulation: the list should be explicit, discussed and defended.	The options will likely be constrained by existing datasets but 'the capabilities that would have been appropriate but for which no information is available' (2003: 70) should also be discussed and reported on.
Methodological justification: clarify and justify the method for putting the list together.	One method could be first, unconstrained brainstorming; second, refinement in light of existing literature and data on the issues of concern; third, engaging with other lists of functionings/capabilities; and fourth, debating the list with other people (Robeyns, 2003)
Sensitivity to context: the level of abstraction should be appropriate to the research objectives.	Functionings relevant to economic or social justice discussions will be less abstract than those relevant to philosophical discussions.
Different levels of generality: This is distinct from but related to the third criterion. It states that 'if the specification aims at an empirical application, or wants to lead to implementable policy proposals, then the list should be drawn up in at least two stages. The first stage can involve drawing up a kind of ''ideal'' list, unconstrained by limitations of data or measurement design, or of socio-economic or political feasibility. The second stage would be drawing up a more pragmatic list which takes such constraints into account. Distinguishing between the ideal and the second-best is important, because constraints might change over time' (2003: 70-71).	For example, including currently unavailable data on care labour on the 'ideal' list contributes to the case for collecting actual data on care labour. The multi-stage procedure guards against the unreflective production of lists which reproduce existing biases.
Exhaustion and non-reduction: listed functionings/capabilities should include all important elements and each element should not be reducible to other elements (Robeyns, 2003).	

Appendix B

Example Research Question: *A reduction in measures of transport poverty is a key element of the just transition. Under different scenarios (T3.2) - to what degree does the switch to electric cars and buses contribute to a decreased (or increased) risk of transport poverty and a significant decrease in emissions*

To note, the findings may not align with expectations i.e. the switch to electric cars – given how policy measures are currently structured – may lead to an increase in transport poverty (it may widen the gap in terms of burden sharing the costs of transition measures) and it may not lead to the expected decrease in overall emissions either – maintaining a car dependent transport system may be increasingly unsustainable.

There is a need to:

- Understand the case study context (e.g. intersections of electric vehicle policies with related policies – e.g. sustainable transport policies, spatial planning policies, employment policies re remote work etc.)
- Identify how the combined policy measures at a systems level produce structurally disadvantaged and structurally advantaged cohorts. A key part of an intersectional analysis is the identification of which cohorts are consistently advantaged through policy measures.
- Map policy measures and governance constraints in line with frameworks and dimensions of justice.
- Compile a list of existing indicators relevant to measuring the risks generated by the impact of combined policy measures at a systems level in each country/region. Data will differ per region/country
- Identify possible severity and likelihood scales of disadvantage and advantage
- Identify and align mitigation measures with frameworks and dimensions of justice. The goal of a design and systems-led approach is to effectively intervene in the system rather than addressing specific issues in isolation from wider systems.
- Design the platform in line with the need to manage conditions of deep uncertainty (rather than trying to remove them) - and enabling actors to develop critical thinking, systems thinking and adaptative capacities

SYSTEMS THINKING FOR TRANSPORT - AVOID-SHIFT-IMPROVE FRAMEWORK
The Avoid-Shift-Improve (ASI) framework for the transport sector aims to develop sustainable transport systems. Studies demonstrated that ASI strategies can account for 40-60% of transport emission reductions, and at lower costs than improve strategies only (SLOCAT, 2021). Initially developed in Germany in the 1990s, the ASI approach includes three hierarchical pillars: first, 'avoid' measures should be addressed to improve the transport system as a whole; second, the 'shift' is brought in to efficiency of individual trips and finally, the 'improve' pillar is implemented with a focus on vehicle and fuel efficiency (TUMI, 2023). An example of such approach includes the following model: • AVOID: Reduce the frequency and distance of car trips. This could be achieved by encouraging remote work and/or digitalisation of services • SHIFT: Move towards more sustainable transport, such as using public transport or active transport (cycling or walking) • IMPROVE: Promote fuel-efficient vehicles, such as EVI. This step should be in addition to the first two, rather than a priority.

However, several issues need to be elaborated in this context. First, as argued by Hampton and Whitmarsh (2023), the reduction in travel is related to spatial planning, employment relations and public services reform and so requires systems-wide, integrated policy design. Secondly, the shift towards remote work or the digitialisation of services may result in social exclusion and further marginalisation, particularly in rural locations. The risks of pursuing any one policy measure, without supporting systems in place, needs to be assessed i.e. there is a need for a whole-system approach, with such solutions as 15-minutes neighbourhoods or prioritizing access to local amenities, such as care facilities (Moreno et al., 2021). A whole systems approach requires inter-departmental efforts, as well as intersectional inequality analysis. The application of the ASI framework needs to be place based and supplemented with additional frameworks as required e.g. Transport-Related Social Exclusion framework (Lucas, 2019). Unintended impacts of systems change must be anticipated and accounted for e.g. the withdrawal of in-person services. These could be addressed through the prioritisation of social innovation measures (e.g. social licensing) in tandem with technological changes. Finally the environmental costs and benefits of policies which seek to achieve social inclusion objectives must be assessed.